

Superdiffusion and nested oscillation in visual search dynamics

Jude Luca Metcalf

This thesis is presented as part of the requirements for the conferral of the degree:

Bachelor of Science (Honours)

Supervisors:
Dr. Brendan John Harris

The University of Sydney
School of Physics

2026

Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

Jude Luca Metcalf

October 19, 2026

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Fusce volutpat lobortis ipsum. Pellentesque et rhoncus turpis. Nam hendrerit ligula eu justo hendrerit eleifend a eu nunc. Suspendisse felis risus, pellentesque at pretium a, imperdiet quis mauris. Pellentesque rutrum, mi sit amet faucibus posuere, neque velit mattis ligula, ac dapibus lectus enim vitae arcu. Aenean cursus, mi quis aliquam dapibus, justo neque posuere ipsum, sit amet pharetra sem nulla non nisi. Phasellus laoreet faucibus metus id convallis. Integer augue sapien, tempus eu aliquet vehicula, ornare ut ligula. Sed ex magna, tempus ut nulla a, pharetra cursus dui. Aenean dapibus dui commodo, pretium nunc gravida, dapibus elit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

Acknowledgements

Contents

Abstract iii

Acknowledgements iv

Introduction 1

1 Survey of the Literature 3
- 1.1 Behavioural data . . . 3
  - 1.1.1 Timescales of neural computation . . . 3
  - 1.1.2 Gaze trajectories . . . 4
  - 1.1.3 Existing models of visual searching dynamics . . . 4
- 1.2 Diffusive processes . . . 5
  - 1.2.1 Brownian motion and normal diffusion . . . 5
  - 1.2.2 Anomalous diffusion . . . 6
  - 1.2.3 Lévy stable motion . . . 7
  - 1.2.4 Nested oscillations . . . 9
- 1.3 Abstract computation . . . 9
  - 1.3.1 Neural sampling . . . 9

2 A Unifying Model 11
- 2.1 Model setup . . . 11
- 2.2 Dynamical-Information adaptation . . . 13
  - 2.2.1 Motivations for the introduction of kernel adaptation . . . 13
  - 2.2.2 Adaptive kernel dynamics as a driver of slow rhythmic exploration . . . 13
- 2.3 Local harmonic oscillations . . . 14
- 2.4 Lévy superdiffusion . . . 15

3 Empirical Validations of Model Predictions 18
- 3.1 Theta rhythmicity of saccades . . . 18
  - 3.1.1 Local rhythmicity of saccades . . . 18
- 3.2 Gamma oscillations during visual fixation . . . 20
  - 3.2.1 Previous treatment as ocular microtremor . . . 20
  - 3.2.2 Analysis of fixational Fourier spectra . . . 20
- 3.3 Search superdiffusion . . . 21
  - 3.3.1 Spectrum of fractional moments . . . 21
- 3.4 Spatiotemporal predictions from the model . . . 21
  - 3.4.1 Saliency-tremor frequency coupling in contrast-dependant search tasks . . . 22
  - 3.4.2 Smooth pursuit predictions and parametric-fitting techniques . . . 22

4 Functional Advantages of Rich Spatiotemporal Searching Dynamics 23
- 4.1 Functional roles of dynamical patterns . . . 23
  - 4.1.1 Information gain . . . 24
  - 4.1.2 Information velocity . . . 25
  - 4.1.3 First passage time . . . 26
  - 4.1.4 Shape parameter . . . 26
  - 4.1.5 Gini coefficient . . . 27
  - 4.1.6 Information–entropy gradient . . . 27
  - 4.1.7 Summary . . . 28
- 4.2 Further extensions of studied dynamics . . . 28
  - 4.2.1 Superdiffusive probabilistic sampling . . . 28

4.2.2 Dynamical goals 28

5 Discussion **30**

A Prior models of visual search **32**

Bibliography **33**

Introduction

The brain is among the most complex systems in science, and one of its most important features is its ability to build a rich understanding of the world from a small, selectively sampled fraction of the available input. Nowhere is this clearer than in vision. In humans, an estimated one in five cortical neurons is devoted to visual processing and control [1, 2], yet for every 1 MB of information reported by the retina, only about 40 bit are consciously processed [3–5]. The visual system must therefore be highly selective, and it achieves this by acting rather than passively receiving. High-acuity vision is confined to the fovea, so to build a useful representation of the visible world, an observer must move their eyes, directing the fovea towards successive locations of interest. These rapid ballistic movements, called saccades, organise periods of relative stillness known as fixations during which the brain gathers visual information, and together they form a sequence of "decisions" about where to look next, made several times every second during searching [6, 7]. Each decision depends on what has already been seen, what is currently in view, and the goals of the observer. Eye movements are thus the external output of a closed perception–action loop in which the brain selects the very data on which it will subsequently make decisions about further perception. This process of decision-making informing further perception is called active vision.

This makes gaze an incredibly valuable observable for understanding the brain. Because the eyes can only be directed at one place at a time, gaze trajectories provide a direct, low-dimensional, continuous, and non-invasive proxy for neural attention, so the spatiotemporal dynamics of attention become direct to measure [CITE]. Gaze is informed by many neural processes that are otherwise difficult to access experimentally: the evaluation of stimulus salience [8–10], working memory for regions already explored [11, 12], top-down goals, and the intrinsic timescales of neural activity [13–16]. Even the low-amplitude movements made during fixation, once dismissed as noise, carry structure reflecting the underlying oculomotor and neural machinery [17–19], and are valuable in gathering information during fixation [CITE].

Visual search, in which an observer scans a scene for a target or simply explores it, is among the most demanding and informative settings for studying basic neural decision-making. The observer must balance the exploration–exploitation trade-off, deciding when to remain on promising regions and when to discover new ones, while avoiding wasteful re-exploration of depleted regions, and must do so efficiently and robustly across many possible tasks. The resulting trajectories are rich and structured, with characteristic spatial statistics, temporal correlations, and preferred timescales. Understanding the mechanisms behind them offers a window not only onto vision, but onto the general strategies the brain uses to explore, sample, and decide under uncertainty [CITE]. Alterations in eye-movement patterns are also associated with a range of neurological and psychiatric conditions, giving these measurements value beyond basic science, towards biomarkers of global brain function that aggregate many regions during visual search [CITE].

The structure of these trajectories also invites a physical description. Gaze can be viewed as a stochastic process which has been aggregated from a complex adaptive system in a critical regime with memory effects to explore a complex dynamical landscape efficiently, with a well-defined purpose and goal of optimising a target search. This viewpoint makes it possible to ask quantitative questions about gaze: how memory effects and adaptive timescales improve dynamical sampling, the structure of heavy-tailed statistics of noise-driven motion, and what enforces characteristic sampling timescales. Methods from statistical physics, nonlinear dynamical systems, and non-equilibrium thermodynamics have already provided new ways to characterise and explain the collective and dynamical properties of neural computation [CITE], and they offer a language for gaze that is both mechanistic and mathematically precise; such a well-defined mathematical model then permits an understanding of the system which drives these dynamics, the attention mechanism of the brain itself. Gaze is also a rare example of a driven, history-dependent stochastic system for which high-resolution data are readily available, making it a natural testbed for such theories. In this thesis, we take inspiration from many exciting novel methods of desigining abstract sampling, such as adding an information-aware

landscape adaptation, utilising heavy-tailed superdiffusive exploration of a landscape, and employing a nested timescales of exploration of a nontrivial landscape.

Existing work has approached gaze dynamics largely one scale at a time, both in terms of space and time. Studies of large-scale saccades characterise their timing, amplitudes, and kinematics [6, 11, 20]. At the scale of whole search trajectories, stochastic models describe global strategies with heavy-tailed, Lévy-like exploration [21–23], or else as through global information-maximising strategies [CITE]. The faster- and shorter scales of fixational eye movements are then treated as a separate subject, through stochastic descriptions of their high-frequency noise statistics [7, 24, 25] or through their perceptual role in rapid information-gathering [26, 27], as well as the low-amplitude drifts during fixation [CITE]. The various timescales of attention more broadly, such as theta rhythmicity, form a further literature rooted in neural recordings and behaviour [13, 14, 16]. Each account is valuable on its own terms, but they are developed with different formalisms, different data, and often different assumptions about what the eye is doing.

Alongside these, the dominant recent approach to modelling human gaze comes from machine learning. A large body of work trains deep networks to predict either gaze trajectories [28–33] or saliency maps indicating where observers are likely to look [9, 10, 34–37]. Most notably, DeepGaze III [38] predicts human fixation locations with high accuracy given the image, the fixation history, and systematic biases in how people search [34, 38, 39]. Such models have been used to test hypotheses about search, showing for example that simple low-level contrast is often sufficient to predict fixations [40], and that late fixations become dominated by intrinsic gaze biases and decouple from image content [38]. Yet these networks are optimised to predict where the next fixation will fall, not why gaze behaves as it does. Their internal representations are difficult to interpret, they do not readily connect to biological mechanisms, and they say little about the temporal organisation of search as a dynamical process [CITE].

Each of these approaches leaves a gap; both between biological realistic accounts and empirical accuracy, and between the fine and coarse scales of perception which should each be captured under a single theory. Saccades and fixations are not independent processes: the regularity, size, and duration of saccades are influenced strongly by information gathered during fixation [CITE]. Models built for a single scale cannot capture this coupling, and their parameters are fitted separately and carry no common physical interpretation. As such, one cannot simply combine such theories into a patchwork theory, as this will not accurately describe the neural control of gaze dynamics. The combined large- and small-scale movements are also rarely measured together, so there are few quantitative constraints on how a unified model should behave across scales [41]. As a result, we lack a single, biologically motivated model in which the heavy-tailed spatial statistics of search, the rhythmic timing of saccades, memory of explored regions, and the structured dynamics of fixation arise from the same underlying biologically-informed mechanism. As such, a unified mechanistic account of search dynamics, going beyond fixation prediction, remains largely unexplored.

Such an account must reconcile the phenomena above while remaining biologically plausible, parsimonious, and capable of generating testable predictions about novel or unexplained aspects of search. It should treat these features as different aspects of one stochastic, history-dependent process rather than as separate phenomena patched together. This thesis aims to unify these behaviours within a single functional, dynamical model of human visual search, linking the statistics of gaze to the principles by which the brain actively samples its environment.

Chapter 1: Survey of the Literature

In this thesis, we aim to unify the rich gaze dynamics found during active visual search under a single physics-inspired model, admitting a deeper understanding of neural computation. There is growing success in turning methods from statistical physics and nonlinear dynamics towards the area of computational neuroscience. In particular, the rich scale-free activity of neural circuits inspires similar analysis to unify the dynamics of gaze trajectories across spatiotemporal scales, and present a physics-inspired dynamical understanding of neural computation in search tasks. In Section 1.2 we introduce the framework of stochastic modelling, Lévy motion, and the associated anomalous superdiffusion which informs the dynamics of the FNS sampler. Section 1.3 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts. Section 1.1 provides the background for much of the behavioural data we present and model in this thesis, and the previous methods used to understand and model the complex dynamics found in gaze trajectories especially.

1.1 Behavioural data

To ground the physical background of this thesis, this section reviews the empirical behavioural data which characterise human visual inference. The visual system does not operate as a passive receiver of sensory input, but rather as an active, dynamic sampler of a changing environment [42]. This active sensing manifests across two coupled physical observables: the internal dynamics of rich neural oscillatory networks, and the external, mechanical execution of gaze trajectories. Crucially, these domains inform and drive each other, and each exhibit complex, scale-free dynamics in their activity. Characterised by background spectra with power law tails and anomalous diffusion, the visual system is capable of devoting attention to many spatiotemporal scales of event. Viewing these biological phenomena through the lens of statistical mechanics provides a rigorous foundation for mathematically modelling the visual system. This section establishes these empirical constraints by first examining the functional role of neural frequency bands and theta-gamma coupling (Section 1.1.1), before detailing the multi-scale, stochastic regimes of human gaze trajectories (Section 1.1.2). Together, these behavioural patterns motivate the fractional dynamics given by the FNS equations of motion, and the developments upon that model in the subsequent sections.

1.1.1 Timescales of neural computation

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands are: delta (0.1–4 Hz), theta (4–8 Hz), alpha (8–14 Hz), beta (14–30 Hz), and gamma (30–220 Hz) [46, 47]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to attention and search tasks of computation. Theta rhythms have been shown to modulate attention during auditory tasks [48, 49], visual search tasks [14, 16, 50], and in analysing the general attentional mechanism of the brain [13, 46, 47]. Intervals between theta peaks are finding stronger interpretation in the neuroscience literature as discretely sampling the environment for integration into the internal model of the brain. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [13, 48], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [14, 16]. Accordingly, the firing patterns which define the lowest level of information carried in the brain are regularly found to be gamma in scale, almost always within 30–100 Hz [45, 51–54]. The coupling between the phase of theta activity and the amplitude of gamma activity is known as theta-gamma coupling (TGC). This phenomenon, a specific form of phase-amplitude coupling (PAC), is schematically represented in Figure 1.1. The evidence and theory behind TGC are described in a systematic review by Ursino and Pirazzini [45]. This thesis provides a computation view of the origin

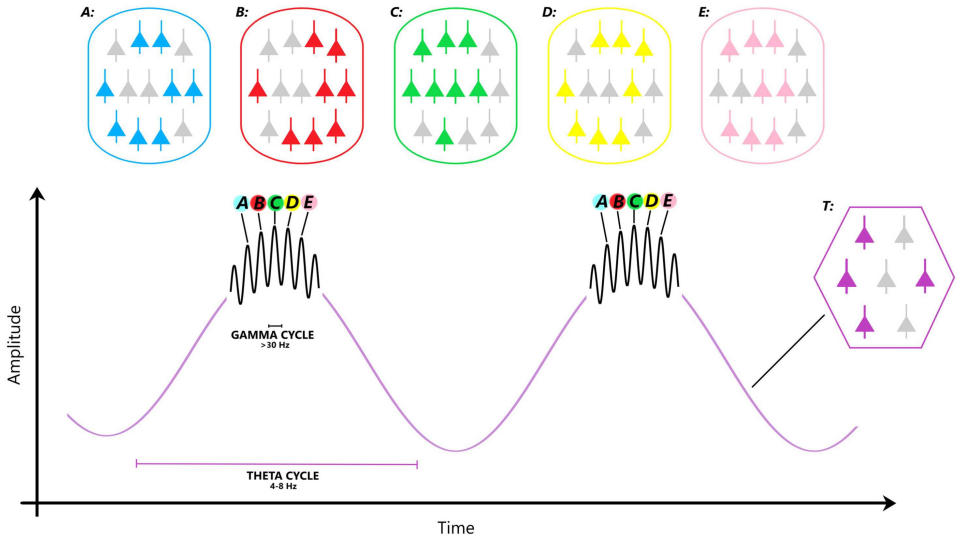


Figure 1.1: A schematic representation of theta-gamma coupling (TGC). Here, neural population patterns $\mathbf{A} - \mathbf{E}$ are activated in sequence on the timescale of a gamma oscillation $f > 30$Hz, triggered by the phase of a modulating theta wave of neuron population **T** on the theta timescale of $4 - 8$Hz. Such a formulation of TGC as correctly ordering signals has its foundations in the work of Lisman and Jensen [43], and was subsequently expanded to describe the dynamic neural mechanism during sensory exploration to modulate close attention upon individual centres of focus [44]. *Schematic adapted from figure 2 of Ursino and Pirazzini [45]*

and function of TGC in the visual system as modulating periods of attention during which fixations exhibit dynamics of active sensing.

1.1.2 Gaze trajectories

Healthy human gaze trajectories consist of different dynamical regimes. The most studied pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as saccades [6, 23, 55, 56]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight. This paradigm has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parameterised by the stable parameter α has found some limited use as a biomarker for ADHD [57]. Broader saccade metrics have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia [58–61], autism [62, 63], OCD [62], bipolar disorder [62, 63], and Parkinson's disease [62, 63]. The dynamics of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [7, 18, 19, 64]. Detailed examinations and systematic reviews of these behaviours have not considered their computational role in active sensing and efficient search. The work presented in this research strongly motivate a closer and more biologically informed understanding of the dynamics of gaze trajectories, across a number of spatial and temporal scales, to diagnose and understand the function of global functioning of the brain, much of which controls the visual system.

1.1.3 Existing models of visual searching dynamics

Progress towards modelling gaze dynamics during visual search spans physics, neuroscience, and machine learning. As such, existing models have been built with very different aims and methods. Cognitive and biologically inspired models explain where attention is drawn through biologically-inspired saliency maps and top-down guidance [148, 149, 151]. Normative models ask what an optimally informative searcher would do, selecting saccades by Bayesian belief updating and entropy reduction

[112]. Physics-inspired models may describe gaze through a dynamical system, through variational least-action principles [21] or through stochastic descriptions of fixational drift [7]. More recently, deep networks trained on eye-tracking data predict human scanpaths on natural images with high accuracy [30, 38].

Each approach captures an important aspect of search, but so-far models must focus on a specific feature and scale of gaze dynamics. Most treat search as a sequence of discrete saccades and neglect continuous-time dynamics; fixational activity is modelled separately from saccades, and the most accurate predictors are not designed to be biologically interpretable. No model among those reviewed combines continuous-time dynamics, saccadic jumps, fixational motion, and an explicit informational account of search. The present model draws on three of these approaches: the information kernel of the ideal Bayesian searcher, the second-order saliency-driven dynamics of the least-action model, and the structured fixational motion of the self-avoiding walk. A full comparison of the most influential and successful models, along with a comparison of their core domains, aims, and key features, is given in Table A.1 in Appendix A.

1.2 Diffusive processes

In formalising empirical observations of human gaze and neural activity into a mathematical model, this thesis relies on the framework of non-equilibrium statistical mechanics and stochastic calculus. When a system is driven by a massive number of microscopic interactions—such as the complex interconnected firing of a neural population—modelling the exact deterministic dynamics becomes intractable. Instead, statistical physics provides the tools to coarse-grain these high-dimensional systems into low-dimensional stochastic differential equations. This section establishes the mathematical progression necessary to model scale-free visual inference. We first review the classical paradigm of normal diffusion and Brownian motion (**??**), which serves as the foundational null model for physical exploration. We then examine how biological systems frequently violate these classical assumptions, leading to anomalous diffusion (Section 1.2.2). Finally, we introduce Lévy stable motion (Section 1.2.3), a class of fractional stochastic processes that mathematically capture the heavy-tailed, intermittent dynamics characteristic of optimal visual search and neural activity.

1.2.1 Brownian motion and normal diffusion

Statistical modelling has been fundamental in understanding complex systems: separating the dynamics of macroscopic observables into deterministic terms and stochastic terms reduces high-dimensional systems to a small number of variables of interest [65, 66]. Across physics, economics, geology, climate science, machine learning, and neuroscience, stochastic processes are often the most natural, tractable, and accurate modelling choice when a system is too complex to model in full. The universality of these stochastic models is explained by the application of the central limit theorem, where the sum of very many small interactions over macroscopic time limits to a Gaussian random variable. This technique turns out to be generalisable to a wide class of coupled non-stationary processes out of equilibrium, as described in Izvekov [67], whose techniques can describe the mean effect of a neuronal spiking patterns as demonstrated in Camera [68]. Such processes thus produce the most ubiquitous stochastic process: the Wiener process W_t, equivalently called Brownian motion, which has stationary and independent Gaussian increments across all non-overlapping time intervals. The Wiener process is a very simple and analytically elegant model of the coarse-grained effect of very many small finite-variance interacting influences. A Wiener process is defined only by its timescale, with a variance over trajectories after time t of t itself. Wiener processes yield time series which are almost surely continuous but nowhere differentiable. The Wiener process is understood as a model of the long-time dynamics of a variable affected by very many small uncorrelated finite variance influences [65]. Classical examples of processes modelled by the Wiener process are: the movement of a macroscopic particle in a non-turbulent fluid [65], the gravitational influence of an orbiting collection of stars on supermassive black holes [69, 70], and (arguably) the movement of low-volatility stock prices [66].

To include random variables such as Brownian motion in dynamical systems, physicists and mathematicians have developed the tools of stochastic calculus. Brownian motion about a potential land-

scape $V(\mathbf{x})$ is often described in dynamical systems contexts in terms of the standard first order Langevin equation:

$$d\mathbf{x}_t = -\gamma \nabla V(\mathbf{x}_t)dt + \sqrt{2\gamma}dW_t \tag{1.1}$$

Where the constant γ describes the noise strength of the Brownian motion. To describe how the probability distribution of this walker $P(\mathbf{x}, t)$ evolves through time, the Fokker-Planck equation is used. In the case of Equation (1.1), the Fokker-Planck equation is:

$$\partial_t P(\mathbf{x}, t) = \frac{\gamma}{2} \nabla \cdot (P(\mathbf{x}, t)\nabla V(\mathbf{x}) + \nabla P(\mathbf{x}, t)) \tag{1.2}$$

whose unique stationary distribution is the Boltzmann-Gibbs distribution:

$$\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})] \tag{1.3}$$

which yields the important relationship, where a potential can be constructed from a normalised target density via $V(\mathbf{x}) = -\log \pi(\mathbf{x})$. Importantly, the dynamical system defined by Equation (1.1) is ergodic, and thus under Equation (1.2), any initial distribution will converge to $\pi(\mathbf{x})$.

Importantly, the normal Fokker-Planck equation 1.2 captures the normal diffusion of a Brownian particle from a known initial condition $\mathbf{x}_0$ as described by the Langevin equation 1.1. The solution to the Fokker-Planck equation from an exact initial condition $P(\mathbf{x}, 0) = \delta^n(\mathbf{x} - \mathbf{x}_0)$ is dominated over short times by the ∇P term as $\nabla P(\mathbf{x}, t) \gg P(\mathbf{x}, t)\nabla V(\mathbf{x})$, reducing the Fokker-Planck equation to the heat equation $\partial_t P(\mathbf{x}, t) = \frac{\gamma}{2}\nabla^2 P(\mathbf{x}, t)$. As such, the leading-order short-time solution is

$$P(\mathbf{x}, \Delta t) \propto \exp\left[-\frac{\|\mathbf{x} - \mathbf{x}_0\|^2}{2(\gamma \Delta t)}\right] \tag{1.4}$$

which demonstrates an interesting behaviour: the short-time scaling of variance in position will be linear, as $\mathrm{Var}[\mathbf{x}_{\Delta t}] = \gamma \Delta t$. Thus γ is often called the timescale of the Brownian motion, as it defines the rate of linear diffusion of variance with time. This linear scaling of the variance of trajectories with time defines normal diffusion, and understanding those systems whose random motion does not reduce to this condition is crucial in understanding stochastic motion.

1.2.2 Anomalous diffusion

The universality of normal diffusion in systems which are driven by Brownian motion as above has been studied closely, most influentially as in the movement of particles in a fluid [70–73]. These dynamics inform the definition of normal diffusion through this variance scaling with time, rigorously calculated as the mean squared displacement (MSD) $\langle \|\Delta \mathbf{x}(\Delta t)\|^2 \rangle \propto \Delta t$. This MSD linear in time defines normal diffusion. Such normal diffusion is universal to any realisation of Brownian motion, and thus any finite-variance aggregated process. To study such diffusive random walks and their long time behaviour the continuous time random walk (CTRW) model was developed, in which a trajectory has subsequent step sizes and durations drawn from finite variance distributions. Such models will universally reproduce this normal diffusive scaling property of $\langle \|\Delta x(t)\| \rangle \propto t$, and as a result of the central limit theorem, will approach Brownian motion. However, increasingly interesting physical systems have been found which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle \|\Delta x(\Delta t)\|^2 \rangle \propto \Delta t^{\lambda_2}, \quad \lambda_2 \neq 1 \tag{1.5}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \lambda_2 < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \lambda_2 < 2$ where diffusion scales faster than normal. The case of $\lambda_2 = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$. An important phenomenon often found in diffusive processes is fractal self-similarity. A Brownian motion is self-similar across all scales, and is nowhere smooth, and in physical systems demonstrating anomalous diffusion, self-similarity resulting from criticality is often reported. Later in Section 1.2.3,

we will find that it is advantageous in the study of anomalous superdiffusion to capture this multi-scale diffusive self-similarity through an extension of the λ_2 scaling of MSD to a spectrum of absolute moments $\lambda(q)$ which describes the diffusive self-similarity which unifies across many scales of study.

Anomalous subdiffusion has been shown to result from heavy-tailed wait-times between interactions [72], such as in trapped fluid flow [74], controlled diffusion through cell membranes [72], and protein mixing [75].

Anomalous superdiffusion

Of great importance in studying the dynamcs of visual search, superdiffusion has been studied in the contexts of complex exploratory biological systems. We model the efficient exploration of gaze dynamics through Lévy superdiffusion, which is the standard dynamical model of superdiffusion in physics and is presented in detail in Section 1.2.3. Such Lévy superdiffusion has a strong link to the problem of designing a strategic search for scarce resources. However the true computational benefit of superdiffusion has become a subject of debate in the literature. On one side, the Lévy foraging hypothesis posits that the standard superdiffusive CTRW, the Lévy walk, is an optimal strategy when searching for sparse and randomly distributed resources. On the other, researchers have proposed that such a model is biologically unrealistic and a result of artificial modelling conditions.

The finding that Lévy superdiffusion is optimal in random search was first reported in a letter to the journal Nature in 1999 by Viswanathan et al. [76]. In the original paper proposing the Lévy foraging hypothesis a Lévy walk with superdiffusion $\langle x(t)^2\rangle \propto \frac{t^2}{\ln(t)}$ on the transition to ballistic diffusion was found to be optimal in a search for sparse resources [76]. Recently, physicists and biologists have questioned if this result is imposed by an unrealistic modelled task, an argument described most strongly just this year by Pyke, Kepert, and Chivers [77], as well as in general analyses of superdiffusion in [78–80]. However, despite such disagreement, the optimality of the superdiffusive Lévy walk has been reproduced in diverse functional search models [81–83], in a population of diverse searching agents with a range of superdiffusive exponents λ_2 [84] as well as in a large study against several different search conditions [81], and have been reported in the empirical search paths of biological organisms from the flights of drosophila flies [83, 85], albatross' searches [86, 87], and strongly across many marine predators [88, 89].

The result of these analyses yields a complex understanding of Lévy superdiffusion, advantages of Lévy walks have been shown to be model-dependent, with the optimal performance of Lévy walks requiring biologically inaccurate depictions of search paths [77, 80, 90]. These arguments marks a split in the treatment of Lévy walks, with groups studying them as informative baseline functional models of abstract search [76, 81, 84, 90], and others led by Graham Pyke presenting this as a misrepresentation of true intelligent organisms [77, 79, 80]. This thesis provides a biologically realistic and well-motivated superdiffusive search for research into a case for optimal Lévy motion. See Reynolds [78] for an in-depth discussion of a richer view of biological Lévy superdiffusion.

1.2.3 Lévy stable motion

To mathematically capture the intermittent, scale-free nature of visual search, we turn to the Lévy flight: a superdiffusive stochastic process that permits close analysis, due to it's very foundational mathematical definition. Here, the Lévy flight L_t^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric α-stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution), where α is called the characteristic exponent and relates to the exponent of the distributions powerlaw tails. $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem, making them ideal to describe a universality class of superdiffusive processes. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law tail of the density function, with the probability density function following $p(x) \propto |x|^{-1-\alpha}$. As a result, the Lévy flight L_t^α after time t is defined by its fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.6}$$

where $\stackrel{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha \to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must generalise the standard Langevin Fokker-Planck equation (1.2) to the Lévy motion using fractional calculus to capture the nonlocal transport of the particle. The details of such an analysis are beyond this thesis, however the fractional Fokker-Planck equation is the tool used to understand the superdiffusion, nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [91]). Many of the simple analytic properties of the Wiener process are no longer true of a Lévy flight. Notably, since these flights have infinite variance in steps sizes, the mean squared displacement diverges. Thus their superdiffusion is better measured using an extension of MSD to fractional absolute displacements, through the definition of the moment spectrum function $\lambda(q)$ as in [92]:

$$\langle \|\Delta x(t)\|^q \rangle \propto t^{\lambda(q)}, \quad q > 0 \tag{1.7}$$

where q is the order of the fractional absolute displacement, and the power law slope will converge for an α-Lévy flight to $\lambda(q) = q/\alpha$ when $q < \alpha$. As such, for Lévy flights with exponents $\alpha > 1$ superdiffusion is well-characterised by the mean absolute displacement, where $q = 1$ in Equation (1.7).

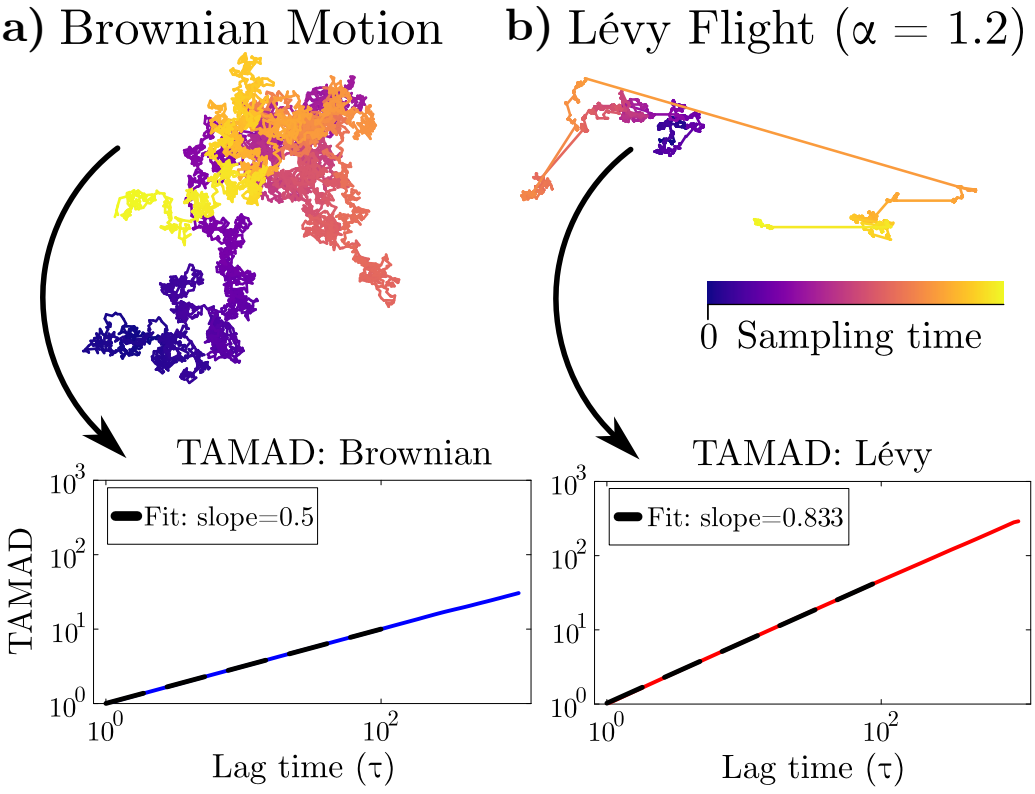


Figure 1.2: Schematic representations of stochastic motion and their respective diffusion. **a)** Schematic representation of a 2D Brownian motion, along with its time-averaged mean absolute displacement (TAMAD), which is the $q = 1$ mean fractional displacement as in Equation (1.7). Normal diffusion is here present as a power law slope of exponent 1/2, giving diffusive exponent $\beta = 1$ as defined in Equation (1.7). **b)** Schematic of a superdiffusive 2D Lévy flight with characteristic exponent of $\alpha = 1.2$, and a MAD slope of 0.833 corresponding to the theoretical superdiffusive coefficient $\beta = 2/\alpha = 5/3$.

Importantly, trajectories of a Lévy flight are no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite à gauche) with respect to time, which describes functions

which are right continuous with well-defined left limits [93]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [94, 95]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [93]. Importantly, it follows from the Lévy-Itô decomposition theorem that a Lévy flight can be decomposed as the sum of two processes [96]:

1. A compound Poisson process of large jumps with $\|\Delta\mathbf{x}\| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number of such jumps in any finite time period.

2. A square-integrable martingale (mean 0) jump process with $\|\Delta\mathbf{x}\| < 1$. This process is then finite-variance, and as such is subject to the central limit theorem, meaning that under temporal coarse graining it approaches a Wiener process.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $|\Delta x| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta x| < 1$. In the context of visual searching, such a division naturally inspires the division between large jumps of attention, and periods of prolonged attention. As such, we are interested in why such superdiffusive scale-free dynamics may have a powerful in computational advantage in exploring an abstract unknown landscape.

1.2.4 Nested oscillations

1.3 Abstract computation

The stochastic dynamical exploration of an information landscape is a powerful paradigm of general computation, and is known as probabilistic computation. Such a landscape in this context is a representation of a target probability distribution. Sampling processes represent and evaluate these distributions through the efficient production of informative stochastic samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods aim to draw sampled independently from a target distribution. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from sparse samples of large multivariate distributions inferred from observational data [97, 98]. This framework has recently gained significant traction as a basic scheme to describe biological computation. The variability in firing patterns of neural populations would thus maintain an internal model of perceptual uncertainty and evolving understanding, giving a functional role to chaotic neural activity [99–101]. This section outlines the paradigm of neural sampling as an interpretation of the variability in neuron firing patterns and visual search gaze trajectories (Section 1.3.1) and how these interpretations fit together with machine learning applications of probabilistic sampling (**??**).

1.3.1 Neural sampling

The framework of neural sampling posits this paradigm that the variability of cortical neural responses is an important functional feature of biological computation. Otherwise regarded as noise, stochastic neural activity patterns are thus presented as samples drawn from a posterior distribution over possible sensory interpretations [102, 103]. The neural sampling hypothesis has been shown to perform and predict network-level empirical patterns of neural activity in the V1, and recently by Terada and Toyoizumi in biologically realistic cortical networks, better than alternative theories [102–105]. Perceptual uncertainty is then encoded by the variability of these cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [97, 102, 103]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of neural computation.

Studies into neural sampling show that variability is not strictly a result of explicit stochastic noise, as traditionally believed. Instead, recent work demonstrates that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models are built to capture [103, 105]. Terada and Toyoizumi link this sampling of a posterior distribution to Bayesian learning through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [105]. Further biologically plausible implementations of such Bayesian sampling are energy-based models (EBMs) which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [106]. The sampling hypothesis has thus given a convincing theory of biological neural computation, with testable and well-supported evidence that neural variability has an important functional role in computation [102, 104]. Traditional Gaussian and Poisson noise models representing these stochastic network dynamics inherently correspond to continuous struggle to capture the full dynamics of complex interconnected inhibitory-excitatory (IE) biological neuronal network [107–109]. Further study into the dynamics of these IE networks has thus motivated further functional models of neural activity.

Visual search

This thesis presents visual search tasks as such a paradigm where a sampling model is well motivated, and driven by the underlying neural sampling process as well. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [9, 110–112]. In the context of visual searching, visually interesting targets with *saliency* as the equivalent of the target landscape, and where the information is located in a target image. Here, saliency is equivalent to the Boltzmann log-likelihood potential from the probability distribution of fixations $p(\mathbf{x})$

$$S(\mathbf{x}) = -\log(p(\mathbf{x})) \tag{1.8}$$

where visual search paths are most likely to be drawn to is a common technique whic [8]. As such, undirected visual searching of an image will show the areas most salient to the brain, allowing insight into the computational sampling mechanism of the visual system. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [113]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [111, 112, 114].

Fractional neural sampling

The FNS dynamics are called fractional, as the non-local dynamics of their distribution require fractional calculus to describe such spatial superdiffusion in Lévy flights [115]. The theory of FNS thus unifies the scale-free nonlocal dynamics of neural firing patterns with the neural sampling framework derived from the ideal Bayesian scheme of active inference. The critical advantage of FNS over other standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—likely a major feature of processed natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [116–119]. Furthermore, the generality of the FNS description permits the application of FNS equations of motion to gaze dynamics, as the motion of the eyes are driven by an IE neural circuit which is expected to keep key features of the FNS model invariant, such as superdiffusion, heavy-tailed statistics, and nonlocal interactions.

Chapter 2: A Unifying Model

Eye movements during visual search have been thought to be a patchwork of distinct behaviours: nonlocal jumps (saccades), periods of fixation, and fine motion within fixations. This chapter presents a biologically motivated model which demonstrates that these behaviours can arise from a single physical process, and to motivate each dynamical component from first principles rather than treating each scale separately. Uniquely, the presented model captures and predicts the rich nested oscillations and superdiffusion in visual searching dynamics; moreover these dynamical signatures of attentional timescales, gamma-frequency (40–100 Hz) sampling, and superdiffusion take origin from network-level neural activity patterns which have been found in the brain [52]. Inspired by such neural dynamics, we model visual search trajectories as a stochastic process driven by travelling activity waves across the surface of the brain, using the theory of fractional neural sampling (FNS) [91, 126].

The chapter is organised as follows. Section 2.1 defines the model based from FNS dynamics before presenting the search task used as the paradigm throughout this work. Section 2.2 introduces the dynamical-informative kernel which is novel to this work, giving its modelling motivations (Section 2.2.1) and the regular information-driven temporal structure of transitions between regions of interest as they are explored through informational kernel deposition (Section 2.2.2). Section 2.3 derives and quantifies the prediction from neural dynamics that localised trajectories produce high-frequency simple harmonic oscillations. Section 2.4 treats the defining characteristic of FNS dynamics, Lévy superdiffusion, deriving the short-time noise-driven scaling of fractional absolute moments and the monoscaling of self-similar superdiffusion which the model presents. Comparison of each of these dynamics with eye-tracking data is deferred to the next chapter.

2.1 Model setup

Having established the individual spatiotemporal signatures of FNS dynamics, we now define a standard computational framework to test these dynamics against the human behavioral data presented in Chapter 3.

Mathematical formulation of gaze dynamics

The dynamical model of gaze dynamics, inspired by and developed from the FNS theory of neural dynamics, models eye movement trajectories as an underdamped Langevin system driven by an adaptive potential landscape $V_t(\mathbf{x})$ by a kernel of information gathering $k(\cdot)$. Let $\mathbf{x}_t$ denote the two-dimensional position vector, $\mathbf{p}_t$ a generalised auxiliary variable analogous to momentum, $\beta > 0$ the coupling strength parameter between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ the noise strength parameter, and $0 < \alpha \leq 2$ the Lévy tail exponent. The driving noise process dL_t^α represents independent symmetric α-stable ($\mathcal{S}\alpha\mathcal{S}$) Lévy fluctuations.

The deterministic drift $b_\alpha(\mathbf{x}_t)$ drives the system toward an invariant target distribution $\pi(\mathbf{x})$. As derived in [91], the exact non-local drift for arbitrary $0 < \alpha < 2$ is formulated in terms of the $(\alpha - 2)$–order Riesz fractional derivative $\mathcal{D}_{\mathbf{x}}^{\alpha-2}$:

$$b_\alpha(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\nabla_{\mathbf{x}} \ln \pi(\mathbf{x})]}{\pi(\mathbf{x})}. \tag{2.1}$$

where these nonlocal fractional dynamics are treated in detail in Section 2.4. However, in this work we adopt the local gradient approximation, which becomes exact as $\alpha \to 2$:

$$b_\alpha(\mathbf{x}) \approx c_\alpha \nabla_{\mathbf{x}} \ln \pi(\mathbf{x}), \qquad c_\alpha = \frac{\Gamma(\alpha - 1)}{\Gamma(\frac{\alpha}{2})^2}. \tag{2.2}$$

Combining these components, the full coupled system for gaze dynamics evaluated throughout this

work is given by:

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \nabla V_t(\mathbf{x})dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \nabla V_t(\mathbf{x})dt, \\ \partial_t V_t(\mathbf{x}) &= \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_t), \qquad V_0(\mathbf{x}) = S(\mathbf{x}) \end{aligned} \tag{2.3}$$

where as α, β, γ are parameters modulating FNS dynamics, c_α corrects the scale of the gradient drift, and τ controls information updates to the dynamical-informative potential $V_t(\mathbf{x})$. The nature and purpose of the dynamical updates $k(\cdot)$ is described in Section 2.2. Importantly, due to the relationship of a Langevin potential to a targeted distribution as in Equation (1.8), we can construct a continuous-time updating internal-state belief distribution of the search target:

$$\pi_t(\mathbf{x}) \propto \exp\left[-V_t(\mathbf{x})\right] \tag{2.4}$$

Empirical protocol from Näher et al.

To validate this modelled computational framework against human behaviour, we emulate the visual foraging task designed by Näher et al. [41], which requires subjects to sequentially scan an array of dots to locate a hidden target. This is done by choosing biologically accurate values for parameters $\alpha, \beta, \gamma, \tau$ and the information gathering scale of $k(\cdot)$. The task itself is encoded in the initial saliency map $V_0(\mathbf{x}) = \log \pi_0(\mathbf{x})$. These parameters and choice of $\pi_0(\mathbf{x})$ are described in the final part of this section.

In the human protocol established by Näher et al. [41], trials commenced with a pre-search fixation period of 2500 to 3500 ms. Following a 100 ms gap, participants were presented with a search array of white dots, each subtending 0.3° of visual angle. Exactly 20% of these dots functioned as true targets, and subjects were allotted a maximum search time of 5000 ms to successfully foveate a target. To investigate the influence of spatial structure, the study compared a REGULAR task—where dots were spaced identically to the centers of an invisible hexagonal tiling—with an IRREGULAR task, which randomly redistributed the same number of dots across the screen to break spatial regularity. Assuming a standard viewing distance of 57.3 cm (where $1° \approx 1$ cm) and a monitor pixel density of $36.22\,\mathrm{px\,cm^{-1}}$, spatial measurements translate directly between pixels and degrees using a scaling factor of $36.22\,\mathrm{px}\,^{\circ -1}$.

Computational model setup & parameter calibration

The empirical conditions from Näher et al. [41] are translated into numerical model parameters using spatial units of pixels (px) and temporal units of seconds (s). The baseline stimulus potential $S(\mathbf{x})$ is constructed on a 2D grid of Gaussian distributions representing quadratic potential wells over a 16×16 square grid (256 wells total), with 20% randomly designated as targets. Based on fixational eye movement distributions, the standard deviation of each well is set to $\sigma_{\text{well}} = 3$ px. The average axial distance between wells is fixed at 110 px ($\approx 3.04°$ visual angle), bridging the dense (2.3°) and intermediate (4.5°) spacings in the REGULAR human search condition.

The baseline parameters for standard simulation runs are calibrated as follows:

- $\alpha = 1.2$: Captures the characteristic search superdiffusion observed in Section 3.3 while maintaining numerical stability.
- $\sigma_k = 80$ px: Defines an adaptive spatial acuity kernel by the Gaussian $k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right)$. This matches the radial decay scale of human visual acuity ($\approx 2.5°$, corresponding to 87.2 px at a 50 cm viewing distance with 4 $\mathrm{px\,mm^{-1}}$ density).
- $\tau = 2 \times 10^{-4}\ \mathrm{s^{-1}}$: Set to yield theta-rhythmic saccadic transitions, as derived in Equation (2.10) and evaluated in Section 3.1. (Note: across simulations, this produces an average ISI of 360 ms, mirroring the empirical saccade rate of 3.53 Hz / 280 ms average ISI).
- $\beta = 1000\ \mathrm{px\,s^{-1}}$ and $\gamma = 1000\ \mathrm{px^2\,s^{-1}}$: Ensures recovery of gamma-rhythmic harmonic oscillations within the underdamped regime for 3 px wells (Section 3.2).

2.2 Dynamical-Information adaptation

The most important component of the dynamics here studied is the dynamical-informative kernel. This component is the most important extension of the work done on fractional neural sampling in works such as Chen and Gong [126] to a full computational model of visual searching during active vision. We call the kernel dynamical-informative, as the kernel influences the model through two coupled processes: the dynamics of the FNS sampler during exploration of a landscape, which is informed by; the information gained during such exploration when considered as an active search. As such, the development of kernel dynamics is informed by previous work in statistical physics and abstract computational sampling, as well as work done informational gathering in the modelled system of visual acuity during search tasks. Importantly, a similar computational process of kernel deposition is called metadynamics in Bayesian probability landscape exploration, see Barducci, Bussi, and Parrinello [127] for the introduction of metadynamics and Bussi and Laio [128] for its developed use as "well-tempered metadynamics" in balancing sampled accuracy and ergodicity. We shall see that these kernel dynamics encode the theta-rhythmic scale of saccades explored in Section 3.1 as well as the rate of information gain explored throughout Chapter 4 when considering functional roles of model dynamics in search tasks abstractly.

2.2.1 Motivations for the introduction of kernel adaptation

The motivations for an addition of a dynamic landscape adaptation of the model are threefold. Firstly, the quantity of information gathered by the gaze trajectory must be accounted for, as this is the fundamental purpose of a visual search. Secondly, the regularity of information-seeking saccades during visual search must be present, most importantly the recently discovered phenomena of strongly theta-rhythmic saccades explored further in Section 3.1, as well as the well-studied phenomenon of inhibition of return (IoR) [11, 12]. Thirdly, any functionally informative description of search dynamics must couple these search dynamics to the current information which has been gathered complete the feedback loop of active vision [8, 110, 129, 130].

To couple gaze dynamics to information gained continuously, we examine the structure of visual information gathering. In this work the target distribution $\pi_t(\mathbf{x})$ will encode the belief in the presence a searched-for target object in 2D space, in a Bayesian sense. Thus, any information during exploration of non-target locations will cause a depletion of belief (probability mass) from the target distribution π locally. This is equivalent in the dynamical model to raising the potential landscape $V_t(\mathbf{x}) = -\ln \pi_t(\mathbf{x})$ locally by some kernel centred at the gaze position whose width represents the spatial acuity of information gathering. This gives a very well-motivated definition of continuous information gathering, as classical information is exactly equated to the logarithm of probability of 'surprise,' which is exactly $\pi_t(\mathbf{x})$ in our search task. Thus, we have a well-motivated instantaneous dynamical law of kernel dynamics as the gaze is positioned at $\mathbf{x}_0$:

$$\partial_t V_{\text{eff}}(\mathbf{x}) = \frac{1}{\tau} k(\mathbf{x} - \mathbf{x}_0) \tag{2.5}$$

where $k(\cdot)$ is assumed to be some approximately radially symmetric kernel function which limits to zero for large $\mathbf{x}$, and is chosen to have scaling $k(\mathbf{0}) = 1$. $k(\cdot)$ will then have a width which correlates directly to the spatial scale of information gathering, which for the human eye is explored here in **??**. Throughout this work, we use an isotropic Gaussian function for $k(\cdot)$, with

$$k(\mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}\|^2}{2\sigma_k^2}\right) \tag{2.6}$$

2.2.2 Adaptive kernel dynamics as a driver of slow rhythmic exploration

We can analytically predict the rate of excursions of an FNS-walker to other modes of the target distribution $\pi(\mathbf{x})$. We consider excursions purely caused by the dynamically adaptive potential kernel controlled by the parameter τ which takes the analogy to the rate of information acquisition defined by some bell-shaped kernel $k(\mathbf{x})$. Based purely on the value of τ, and the approximate shape of $k(\mathbf{x})$ we

can construct a tractable transition condition as follows: The effective potential well in log-probability space becomes concave-down. Such a condition will no longer permit localisation of the FNS walker, and Lévy-driven noise will create excursions to other modes.

We analyse this condition by considering a single isotropic radial Gaussian target distribution centred at zero with standard deviation σ_{fix}. We simplify this problem to simply predict the average interval between excursions within a very localised potential, with excursions largely driven by deterministic kernel dynamics. It is a well known fact that Lévy noise will also introduce Poisson-distributed waiting times between frequent excursions, however analysis of this is beyond the scope of this project. This corresponds to a quadratic potential well in log-probability space. This choice is maximally general, as all local modes will be approximated to second order by a quadratic potential well, using Laplace's approximation. We also consider an adaptive kernel $k(\mathbf{x})$ which is a radial Gaussian in log-probability space with a standard deviation σ_k. For similar reasons as the choice of potential well, such a choice yields the same results for different choices of $k(\mathbf{x})$ which have the same radial curvature. Thus, using a radial coordinate r which is centred at the mode $r = 0$, assuming that $\sigma_k \gg \sigma_{\text{fix}}$ and that the walker is localised at $r \approx 0$, our effective potential function $S(r;t)$ will evolve as

$$\frac{\partial S(r;t)}{\partial t} = \frac{1}{\tau}k(\mathbf{x}) = \frac{1}{\tau}\exp\left(-\frac{r^2}{2\sigma_k^2}\right), \tag{2.7}$$

$$S(r;0) = -\ln\pi(\mathbf{x}) = \frac{r^2}{2\sigma_{\text{fix}}^2} \tag{2.8}$$

after integrating, this yields the form effective potential after time t

$$S(r;t) = \frac{r^2}{2\sigma_{\text{fix}}^2} + \frac{t}{\tau}\exp\left\{-\frac{r^2}{2\sigma_k^2}\right\} \tag{2.9}$$

Then, our excursion criterion is that at time t_θ at the fixation $\|x\| = 0$, the radial second derivative will become negative, leaving the walker unconstrained. The second radial derivative is here

$$\frac{\partial^2 S(r;t)}{\partial r^2} = \frac{1}{\sigma_{\text{fix}}^2} + \frac{t}{\tau}\left[\frac{r^2}{\sigma_k^4} - \frac{1}{\sigma_k^2}\exp\left\{-\frac{r^2}{2\sigma_k^2}\right\}\right]$$

and we find where this second derivative is zero at the fixation point $r = 0$ to find the timescale t_θ

$$\frac{1}{\sigma_{\text{fix}}^2} + \frac{t_\theta}{\tau}\left[0 - \frac{1}{\sigma_k^2}\right] = 0$$

$$t_\theta = \tau\frac{\sigma_k^2}{\sigma_{\text{fix}}^2} \tag{2.10}$$

It is unsurprising that t_θ is proportional to τ. The fact that t_θ is proportional to the squared ratio of the width of the adaptive kernel σ_k and the width of the mode σ_{fix} is of interest. This result shows that the average amount of time dwelt within a potential well will be inversely proportional to the square of the largest variance.

2.3 Local harmonic oscillations

One predictive feature of FNS dynamics is the local oscillations in sampling within confined potentials. We can quantify such high frequency oscillations under FNS dynamics by considering the deterministic part of our FNS equations of motion in Equation (2.3). Considering a Gaussian probability distribution of fixation positions $\pi(\mathbf{x}) \propto \exp\left\{-\frac{\|\mathbf{x}\|^2}{2\sigma_{\text{fix}}^2}\right\}$ and thus logarithmic gradients $\nabla \ln\pi(\mathbf{x}) = -\frac{\mathbf{x}}{\sigma_{\text{fix}}^2}$. Using this as our saliency landscape $\pi(\mathbf{x})$ without considering perturbation to the explored potential from

the slowly updating kernel dynamics, our equations of motion are

$$\begin{aligned} d\mathbf{x}_t &= -\gamma c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= -\beta c_\alpha \frac{\mathbf{x}_t}{\sigma_{\text{fix}}^2} dt \end{aligned} \tag{2.11}$$

where we ignore the stochastic noise driving this equation to only consider the coupled ODE which describes deterministic part of $\mathbf{x} = \mathbf{x}_t$ and $\mathbf{p} = \mathbf{p}_t$, where the loss of subscript in typical when moving from an SDE to an ODE

$$\begin{aligned} \frac{d\mathbf{x}}{dt} &= -\gamma c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} + \beta \mathbf{p}, \\ \frac{d\mathbf{p}}{dt} &= -\beta c_\alpha \frac{\mathbf{x}}{\sigma_{\text{fix}}^2} \end{aligned} \tag{2.12}$$

where the ODE describing $\mathbf{x}$ can be differentiated, and $\frac{d\mathbf{p}}{dt}$ substituted to give one second order ODE for $\mathbf{x}$

$$\frac{d^2\mathbf{x}}{dt^2} = -\frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2} \frac{d\mathbf{x}}{dt} - \frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2} \mathbf{x} \tag{2.13}$$

Solving the characteristic equation associated with this ODE, $\lambda^2 + \frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2}\lambda + \frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2} = 0$, we have

$$\lambda = \frac{1}{2}\left(-\frac{\gamma c_\alpha}{\sigma_{\text{fix}}^2} \pm \sqrt{\frac{\gamma^2 c_\alpha^2}{\sigma_{\text{fix}}^4} - 4\frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2}} \right) \tag{2.14}$$

Assuming an underdamped regime where the discriminant is negative, the resonant frequency ω of the fixational dynamics corresponds to the' imaginary part of the eigenvalues

$$\omega = \sqrt{\frac{\beta^2 c_\alpha}{\sigma_{\text{fix}}^2} - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^4}} \tag{2.15}$$

which yields ordinary frequency

$$f = \frac{1}{2\pi\sigma_{\text{fix}}} \sqrt{\beta^2 c_\alpha - \frac{\gamma^2 c_\alpha^2}{4\sigma_{\text{fix}}^2}} \tag{2.16}$$

2.4 Lévy superdiffusion

The equations of motion for our system are derived from the over-damped Langevin regime, driven by Lévy noise. Such underlying dynamics are known as Lévy flight (see 1.2.3) and are the paradigm superdiffusive system. However, both the adaptive kernel dynamics, and auxiliary momentum dynamics add complexity to the superdiffusive properties of the model. Quantifying superdiffusion in this system exploring a confined and dynamic potential landscape introduces two distinct and challenging problems: firstly, a kind of inter-modal superdiffusion is introduced through adaptive kernel dynamics which expel the walker from potential wells, and secondly a superdiffusion on small spatiotemporal scales results from near-ballistic momentum-driven harmonic oscillations.

Analysis of Lévy flight superdiffusion from the fractional Fokker-Planck equation

Lévy processes are the canonical form of superdiffusion, yet the analysis of the precise form of statistical diffusion scaling requires close analysis. Lévy flight superdiffusion is analysed from the deterministic dynamics of the probability distribution of jump size $P(\|\Delta x\|)$ dynamics of the probability distribution. This is because the dynamics of the probability distribution is governed by the fractional Fokker Planck equation as in [91, 131, 132]. Considering the phase space of our coupled differential equations as $\mathbf{z} = [\mathbf{x}, \mathbf{p}]$, the fractional Fokker-Planck equation can be analysed:

$$\frac{\partial P(\mathbf{z}, t)}{\partial t} = -(\mathbf{D} + \mathbf{Q}) \frac{\partial}{\partial \mathbf{z}} \left[b(\mathbf{z}) P(\mathbf{z}, t) \right] - \mathbf{D} \mathcal{D}_{\mathbf{z}}^{\alpha} P(\mathbf{z}, t) \tag{2.17}$$

Where $\mathcal{D}_{\mathbf{z}}^{\alpha}$ represents the α-fractional Riesz derivative, which generalises the operation of the Laplacian to fractional orders, and is defined in Fourier space as $\mathcal{D}_{\mathbf{z}}^{\alpha}[f] = \mathcal{F}^{-1}\|\mathbf{k}\|^{\alpha}\mathcal{F}[f]$, with the case of $\alpha = 2$ restoring the ordinary Fokker-Planck equation with $\mathcal{D}_{\mathbf{z}}^{\alpha=2} = -\nabla_{\mathbf{z}}^2$. The $b(\mathbf{z})$ drift term is presented in Equation (2.1), and it's local approximation used throughout this thesis in Equation (2.2). The matrices in Equation (2.17) encode the form of the FNS dynamics given in **??**, where $\mathbf{D} = \begin{bmatrix}\gamma & 0\\ 0 & 0\end{bmatrix}$ and $\mathbf{Q} = \begin{bmatrix}0 & -\beta\\ \beta & 0\end{bmatrix}$ are the generalised matrices defining Langevin dynamical systems. $\mathbf{D}$ is the positive semi-definite matrix which defines the diffusion strength, while $\mathbf{Q}$ is the skew-symmetric matrix which defines the coupling between dynamical variables. Solutions of the dynamics in Equation (2.17) are nontrivial, as the fractional derivative results in a nonlinear multivariate integro-differential equation which is difficult to propagate, even numerically. This further motivates our method of analysing individual numerically integrated trajectories of **??**. When considering diffusion over short timescales, the physics of Langevin systems are largely noise dominated. In our Lévy driven system, short times are dominated by α-stable dL_t^{α} Lévy noise, across power-law distributed spatial scales of short jumps and nonlocal jumps to distant positions in short times. This is corroborated by the scaling of Lévy noise as $dL_{\Delta t}^{\alpha} = \Delta t^{1/\alpha} dL_1^{\alpha}$ in Equation (1.6) is stronger than ballistic effects which scale in short times as Δt. Thus, considering the fractional Fokker-Planck equation evolving from a delta function initial condition $P(\mathbf{z}, 0) = \delta^n(\mathbf{x} - \mathbf{x}_0)\delta^n(\mathbf{p} - \mathbf{p}_0)$, and constructing a leading-order short time solution up to $t = \Delta t$ by marginalising the auxiliary momentum $\mathbf{p}$ and using the position coordinate $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}_0$, we have:

$$\partial_t P(\Delta\mathbf{x}, t) = -\gamma\mathcal{D}_{\mathbf{x}}^{\alpha} P(\Delta\mathbf{x}, \Delta t) - \nabla_{\mathbf{x}} \cdot \left[\left(\gamma\mathbf{b}(\Delta\mathbf{x}) + \beta\,\mathbb{E}[\mathbf{p}\,|\,\Delta\mathbf{x}, \Delta t]\right) P(\Delta\mathbf{x}, \Delta t)\right] \tag{2.18}$$

where we can truncate these terms to highest order using

$$\mathbf{b}(\Delta\mathbf{x}) = \mathbf{b}_0 + \mathcal{O}(\Delta\mathbf{x}), \qquad \mathbb{E}[\mathbf{p}|\Delta\mathbf{x}, \Delta t] = \mathbf{p}_0 + \mathcal{O}(\Delta t), \qquad \mathbf{v}_0 = \gamma\,\mathbf{b}_0 + \beta\,\mathbf{p}_0 \tag{2.19}$$

which gives

$$\begin{aligned}
\partial_t P(\Delta\mathbf{x}, \Delta t) &= -\gamma\mathcal{D}_{\mathbf{x}}^{\alpha} P - \nabla_{\mathbf{x}} \cdot \left[\left(\mathbf{v}_0 + \mathcal{O}(\Delta\mathbf{x}) + \mathcal{O}(\Delta t)\right) P\right]\\
\partial_t P(\Delta\mathbf{x}, \Delta t) &= -\gamma\mathcal{D}_{\mathbf{x}}^{\alpha} P - \mathbf{v}_0 \cdot \nabla_{\mathbf{x}} P + \left[\mathcal{O}(\Delta t^0) + \mathcal{O}(\Delta t^{1-1/\alpha})\right] \cdot P\\
\partial_t \hat{P}(\mathbf{k}, \Delta t) &= -\gamma\|\mathbf{k}\|^{\alpha}\hat{P} - i\,\mathbf{v}_0 \cdot \mathbf{k}\,\hat{P} + \mathcal{O}(\Delta t^0) \qquad (2.20)\\
\implies \hat{P}(\mathbf{k}, \Delta t) &= \exp[-\gamma\|\mathbf{k}\|^{\alpha}\Delta t - i\mathbf{v}_0 \cdot \mathbf{k}\,\Delta t]\,[1 + \mathcal{O}(\Delta t)] \qquad (2.21)
\end{aligned}$$

Showing that, to first order in Δt, the fractional Fokker-Planck under FNS dynamics in Equation (2.3) gives a stable distribution

Quantification of Lévy superdiffusion monoscaling

In calculating absolute fractional moments, we can reduce our

$$\langle|\Delta x|^q\rangle = C_q \int_0^{\infty} \left[1 - \mathrm{Re}\left(\hat{P}(k, \Delta t)\right)\right] k^{-1-q} dk \tag{2.22}$$

$$\langle|\Delta x|^q\rangle = C_q \int_0^{\infty} \left[1 - \cos(v_0 k\,\Delta t) e^{-\gamma\Delta t\, k^{\alpha}}\,[1 + \mathcal{O}(\Delta t)]\right] k^{-1-q} dk \tag{2.23}$$

$$\tag{2.24}$$

with

$$C_q = \frac{q(1-q)}{\cos(\frac{q\pi}{2})\Gamma(-q)} \tag{2.25}$$

for $1 < q < \alpha$ and $q < 2$:

$$\langle|\Delta x|^q\rangle = C_q \left[\frac{\Gamma\left(1 - \frac{q}{\alpha}\right)}{q}(\gamma\Delta t)^{q/\alpha} + \frac{\Gamma\left(\frac{2-q}{\alpha}\right)}{2\alpha} v_0^2\,\gamma^{-2}\,(\gamma\Delta t)^{\,2-\frac{2-q}{\alpha}} + \mathcal{O}(\Delta t^{1+q/\alpha})\right] \tag{2.26}$$

This analytically gives the prediction that a spectrum of fractional absolute moments will scale as $\langle \|\Delta x\|^q \rangle \propto \Delta t^{\lambda(q)}$ with $\lambda(q) = \frac{q}{\alpha}$. Thus, FNS dynamics are shown to give the same diffusion for a Lévy flight with $q < \alpha$, and $\lambda(q) = 1$ for $q > \alpha$, with an oscillatory diffusion to higher order due to v_0 momentum-driven terms. This piecewise superdiffusion is a result of the discontinuities in jump dynamics of a Lévy flight which lead to divergent moments, and may lead analysis to present Lévy flight behaviour as being a multifractal, despite it's monofractal scaling definition. There are a number of modelling choices which can be made to remedy this broken scaling law by taming Lévy discontinuities. Suggestions for describing a fuller monofractal superdiffusion would be to use a Lévy walk CRTW structure, as described in **??**, or otherwise impose an online low-pass filter to gaze trajectories, simulating more realistic muscular control of the eyes without overly altering the neural dynamical system.

Chapter 3: Empirical Validations of Model Predictions

Our model is designed to be compared against empirical data and against established accounts of the phenomenon it describes, rich gaze dynamics during visual search. In this chapter, we compare each of the modelled dynamics of Chapter 2 with high precision eye-tracking data gathered by collaborators in Näher et al. [41] as described in Section 2.1. gaze dynamics during visual search show many broad and often task-dependent spatiotemporal patterns, so we ask which are most important for understanding computation in the brain. The literature inspires our partition of modelled dynamics, namely 2.2 information rhythmicity, 2.3 local short-time oscillatory sampling, and 2.4 global spatial superdiffusion, maps onto important visual dynamical signatures that have been observed elsewhere. These correspond to (1) theta-rhythmic macrosaccades, (2) gamma-scale 40–100 Hz fixational ocular drift, and (3) global saccadic superdiffusion to an informative visual scene [26, 133]. Importantly, our unifying model provides an understand of each of these dynamics as a function of network-level neural activity, including the novel connection between microtremor and the neural gamma signal, theta-gamma coupling, and the monoscaling of Lévy superdiffusion across spatial scales.

Section 3.1 examines theta rhythmicity in saccades, comparing the model's transitional rhythmicity with the saccade timing in the data. Section 3.2 examines gamma-scale oscillations during visual fixations, testing the model's local high-frequency harmonic oscillations against fixational drift. Section 3.3 examines superdiffusion in search, assessing whether the global spatial statistics of gaze are consistent with Lévy superdiffusion. Finally, Section 3.4 presents the spatiotemporal predictions of the model across these scales and discusses how they unify the experimental and modelling findings above.

3.1 Theta rhythmicity of saccades

The approximate 200 ms timescale of human saccades has long existed experimentally [134–137]. A direct measure of saccade rhythmicity is the direct measurement of the spread of the ISI distribution, assuming a large lognormal spread of intervals Δt. This can be used to construct a measure called the Rhythmicity Index (RI):

$$\mathrm{RI} = \frac{1}{\mathrm{CV}_{\ln}}, \qquad \mathrm{CV}_{\ln} = \sqrt{e^{\sigma_{\ln}^2} - 1} \tag{3.1}$$

where $\sigma_{\ln}$ is the standard deviation of the distribution of ISIs in log-space $\ln \Delta t$, with $\mathrm{CV}_{\ln}$ often called the geometric CV. This RI is an effective measure of the spread of a strictly positive distribution. A Poisson distributed variable will give RI = 1, whereas it is found that for the ISI distribution of subjects considered here for a random NOISE condition is RI = 2.69 [41].

3.1.1 Local rhythmicity of saccades

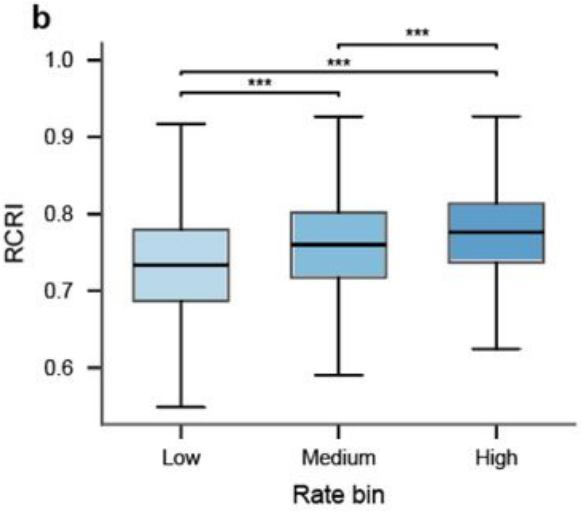
The recent discovery of the strength of saccade rhythmicity by Näher et al. [41] and others [15] required a deep understanding of the rhythmicity of biological systems. A challenge that is often present in biological systems is that timescales are not driven by a perfect clock, but instead the characteristic timescales of certain regular patterns in the brain. As such, the 200 ms estimate for saccades during a visual search is not expected to be a universal timescale across subjects and trials. Instead, the model used for such processes in neuroscientific works is a noisy rise-to-threshold model, where a random walk is constructed, and when it reaches a goal, a decision is made [41] [CITE]. The noise-driven rise for this process is expected to have a rate which is slowly varying about a characteristic timescale, in the case of saccades around 2–6 Hz. Since this slowly varying rate is expected to be stable for a single individual and a single trial, we consider a local measure of rhythmicity for the intervals between the saccades within a timseries, called the Inter-Saccadic-Intervals (ISI). Such a metric can

be proposed—as developed by [41]—with the local Rate-Corrected Rhythmicity Index (RCRI):

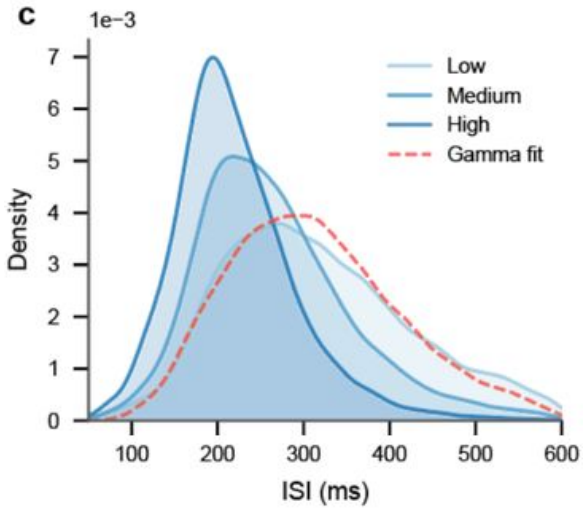
$$\mathrm{RCRI} = \frac{1}{1 + \langle \mathrm{CV}_2 \rangle}, \qquad \mathrm{CV}_2(i) = 2\frac{\|\Delta t_{i+1} - \Delta t_i\|}{\Delta t_{i+1} + \Delta t_i} \tag{3.2}$$

which allows quantification of purely local rhythmicity, comparing each interval Δt_{i+1} to it's predecessor Δt_i, and averaging across all intervals considered i, usually a single subject. The results of such an analysis have resulted in two key conclusions. Firstly, human saccades are very strongly rhythmic, with RCRI ≈ 0.76 for the subjects as described in Section 2.1 depending on condition. Secondly, it has been found that the instantaneous saccade rate is a strong predictor for the local saccade rhythmicity RCRI, with a correlation calculated at $\rho = 0.39, P = 0.001$, with saccade rate found to be a significant predictor of RCRI with $\beta = 0.0114\,\mathrm{Hz}^{-1}, P = 0.014$

those particular individuals and trials binned with a higher saccade rate correlate with higher RCRI and thus stronger comparable rhythmicity [41], as shown in Figure 3.1a.



(a) Rate-corrected rhythmicity index (RCRI) is positively correlated with saccade rate. Collaborators Näher et al. [41] uncovered this statistically significant result, which implies that shorter inter-saccadic intervals (ISIs) will result in more regular ISIs, which corresponds to stronger kernel dynamics as defined in Section 2.2.2.



(b) The probability distribution of inter-saccadic intervals (ISIs) are tightly distributed in a Gamma-10 distribution. Our collaborators in Näher et al. [41] find that saccades are more locally rhythmic than previously considered. In particular, the ISI distribution is a tight Gamma-10 distribution, but the ISI timeseries is strongly locally rhythmic, as captured by the rate-corrected rhythmicity index (RCRI).

Figure 3.1

Interpretations of local rhythmicity and saccade rate for a model of gaze dynamics

These results support our unified stochastic-informative formulation of gaze dynamics. We have shown good agreement between the tight distribution of modelled ISIs, more closely locally. We have also introduced a natural parameter τ with which to modulate the slowly varying theta-scale saccade rate. This also connects the dynamics of information gathering with the dynamics of regular attention in a search task. The results that rhythmicity is strongest where a task contains nearly identical searched objects is encouraging for this information-driven saccade rate theory. A correlation between high saccade rate and high rhythmicity is predicted from the origin of regular dynamical-informational transitions as described in Section 2.2, where a stronger kernel produces more rhythmic transitions, and a weaker kernel will approach an arrhythmic Poisson point distribution of ISIs, reducing the RCRI, as predicted.

3.2 Gamma oscillations during visual fixation

The second most important time- and spatial scale of gaze dynamics are those of high frequency fixational dynamics. Typically fixational dynamics are divided into microsaccades, low-frequency linear drift, and high frequency ocular tremor. There is evidence that microsaccades are driven by the same statistics and process as regular (called macro-) saccades, and linear drifts have been found to be important in gathering information. An account of low-frequency linear drift easily fits into the model presented here, for instance as steep peak in the dynamical kernel or a simple drift in the potential landscape, producing a self-avoiding walk as described in [7] and described in Table A.1.

The current study is the first to make a well-motivated prediction concerning the neural origin and structure of high-frequency ocular microtremor. Identifying ocular tremor as high frequency gamma-scale visual sampling is a novel and useful approach in understanding perceptual sampling. Ocular tremor, under this hypothesis, will originate in the neural control of visual information sampling, likely in the brainstem where eye movements originate [CITE].

3.2.1 Previous treatment as ocular microtremor

The high frequency ocular tremor is the smallest fixational motion, only existing with amplitude of around one minute of arc. Although, it is nonetheless recently shown to be very important in dynamically sampling high contrast details and high spatial frequencies in images. and is increasingly studied as measurement equipment are non-invasive and designed to accurately and measure low-amplitude high-frequency ocular signals.

3.2.2 Analysis of fixational Fourier spectra

This is the first study to examine the high frequency fixational tremor during visual search. Other studies have constructed artificial tasks where subjects are instructed to fixate on a point of interest for extended periods, often stabilising the head with bite-bars. Such setups are not designed to represent natural gaze dynamics, especially during a search task. The computational difficulty in extracting such low-amplitude, high frequency signals from the data in [41] is twofold: Firstly, the fixations only contain an average of around 200 ms of data, and leaving only an approximate 10 periods of such high frequency signals, and secondly such high frequency signals are often error-prone when examining noise-containing positional data.

We are lucky in that we have a large amount of data with a very high quality, sampled at a temporal sampling frequency of 1000 Hz, and with arcsecond precision [CITE PROPER VALUE]. Since the data will then contain low frequency drifts, we expect a power spectral density to scale as $\mathrm{PSD}(\mathbf{x}) \sim f^{-2}$, and thus when analysing, we correct this slope by applying a finite-differencing time derivative to extract velocity, where the null hypothesis of no significant tremor peak would result in white noise $\mathrm{PSD}(\mathbf{v}) \sim 1$.

Our data pipeline here is thus to apply a light 4th-order Buterworth low-pass filter at 200 Hz to handle high-frequency instrumental noise, extract 2D velocity for each eye through central-differencing the positional data, and performing Fourier analysis on such velocity data.

3.3 Search superdiffusion

Superdiffusion is often a prominent pattern in natural dynamical systems, where its existence implies some self-similar fractal structure. Such dynamics have not been the focus of other studies, and often visual trajectories have been considered as a fixational Brownian motion externally interrupted with saccades. We here present novel multiscale superdiffusive pattern in the diffusion of eye movements from eye trajectory data. Many studies which have described eye movement dynamics as normally diffusive stems from the sensitivity of data processing choices in finding anomalous diffusion exploring confined boundaries such as fixations. Conventional analyses often inadvertently obscure critical spatiotemporal features.

The most important methodological choice in accurately characterising visual diffusion patterns is the choice to time-average displacements, and choosing a short timescale over which diffusion scaling is considered. As demonstrated by Vilk et al. [92], aggregating data across disparate temporal regimes can wash out localised scaling behaviours, yielding an artefact of normal diffusion. Studies which construct regimes of normal and subdiffusive regimes are thus common when examining eye movements, but this ignores the continuous-time superdiffusion which we unify under multiple scales. For such studies, see [23, 138–140]

Analysis of multiscale superdiffusion in eye-tracking data

We analyse the binocular-averaged gaze trajectories over all 22 human subjects provided from the study in Näher et al. [41], as described in Section 2.1. From the onset of the search task, we time-average the fractional absolute displacements $\langle \|\Delta\mathbf{x}(\Delta t)\|^q \rangle$ for 30 values of q linearly separated from $q = 0.1$ to $q = 3.0$. These fractional absolute displacements are evaluated across all valid time lags $\Delta t < 1000\,\text{ms}$ with a resolution of $\Delta t = 1\,\text{ms}$. All intervals containing NaN values (such as blinks) are ignored. We then ensemble-average these displacements across all subjects.

We perform a linear best-fit in log-log space of these absolute fractional displacements to find power-law diffusive scaling with time, thus sampling from our moment spectrum function $\lambda(q)$ as defined in Equation (1.7). Each of these fractional absolute displacements are shown in Figure 3.2a. The moment spectrum function $\lambda(q)$ thus is found to have a good fit to $\lambda(q) = (0.893 \pm 0.006)q$ with an error in the fit of $R^2 = 0.999$. This behaviour gives monoscaling across the shortest to longest spatial scales of fixations to saccades. The corresponding Lévy exponent given such linear monoscaling in $\lambda(q)$ is thus found to be $\alpha = \frac{q}{\lambda(q)} = 1.120 \pm 0.007$.

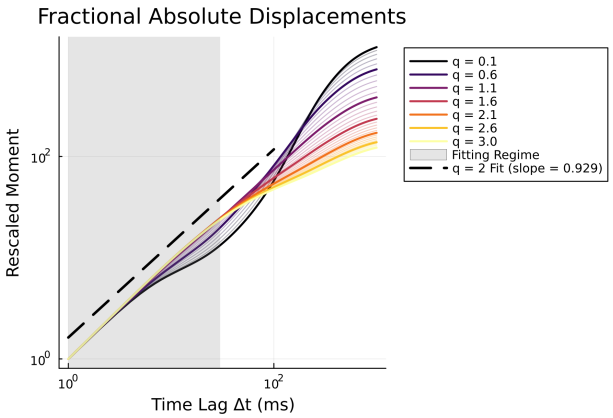
3.3.1 Spectrum of fractional moments

Implications of monoscaling superdiffusion

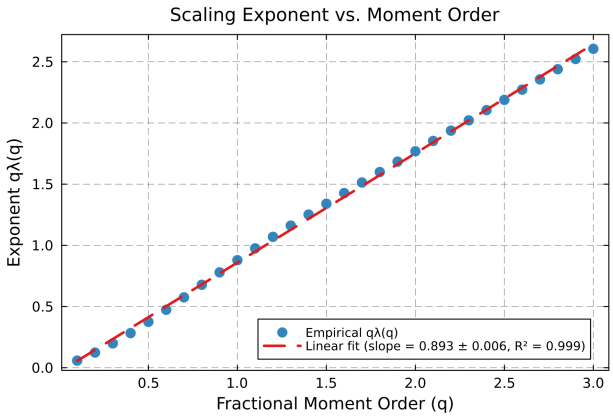
The most important implication of this particular superdiffusion results is that the strong monoscaling of visual search trajectories with Lévy index $\alpha = 1.120$ is robustly multiscale. The robust linear scaling of $\lambda(q) = \frac{q}{\alpha} = \frac{q}{1.120}$ across a spectrum of q-values $0 < q < 3$ is important, as it presents an identical process driving both fixational eye movements — including tremor, drifts, and microsaccades — with small relative values of $\|\Delta\mathbf{x}\|$ emphasised with low values of q, as well as large nonlocal saccades with large relative values of $\|\Delta\mathbf{x}\|$, emphasised through large values of q. Importantly, this validates our construction of a continuous-time unified model of eye movements, unified across all spatiotemporal scales of visual search.

3.4 Spatiotemporal predictions from the model

The above have been analyses of data which already exist in the literature to varying degrees of novelty. In particular, the approximate timescale of saccades has been known for the history of eye movement research, whereas the significance of the rate-corrected local rhythmicity is a very recent result only discovered in the year of this present work, 2026. Ocular microtremor as a phenomenon has been found in many studies, however consensus has never been reached as to it's importance, and it has never been linked to a computational role of gamma-scale rhythms. Lastly, the close mathematical analysis of search superdiffusion has only been recently undertaken, and where it has been studied it has been



(a) The ensemble-averaged spectrum of time-averaged fractional absolute displacements of human visual search trajectories. The fractional absolute displacement generalises the mean squared displacement across a fractional spectrum q. As such, fractional absolute displacement is defined as $\langle \|\Delta \mathbf{x}\|^q \rangle$, and is here rescaled to be plotted as $M_q(\Delta t) \propto \langle \|\Delta \mathbf{x}\|^q \rangle^{\frac{1}{q}}$ with $M_q(1\text{ ms}) = 1$ for the sake of comparison. Time-averaging was done over all 22 subjects from the study conducted by Näher et al. [41], corresponding to 10,000 seconds of data (EXACT SAMPLE SIZE TO BE UPDATED).



(b) The uniform superdiffusive scaling of fractional absolute displacements in human data. The uniform scaling of fractional moments from Figure 3.2a up to short time $\Delta t = 30$ ms gives a very good linear fit of the diffusive exponent $\lambda(q) \approx (0.893 \pm 0.006)q$. This suggests a Lévy stable parameter of $\alpha = \frac{1}{\lambda} = 1.12$. This result demonstrates very strong superdiffusion.

Figure 3.2

done naively as a simple Lévy walk with no awareness of the non-equilibrium potential landscape being explored. However, another benefit of comparing the generative unifying model presented in Chapter 2 to the data is that further dynamical predictions outside of the present literature can be made. Thus, we present experimental paradigms and features of this model which are being developed to better understand visual searching.

3.4.1 Saliency-tremor frequency coupling in contrast-dependant search tasks

That process which has been called ocular microtremor has been hypothesised to be harmonic oscillations in the control of the path of the eye during fixation tasks. An experiment may be designed in which targets of various degrees of contrast are presented to be searched. High contrast, localised points, such as high spatial frequency Gabor patches, would correlate with localised saliency wells. Whereas, low contrast, smeared low spatial frequency patches would correlate with delocalised saliency wells. High contrast wells would correspond to shorter theta timescales, quadratic in the effective saliency well size as in Equation (2.10), and a microtremor rhythm which is approximately inverse to the effective saliency well size as in Equation (2.15).

3.4.2 Smooth pursuit predictions and parametric-fitting techniques

Another experimental paradigm which provides insight into the behaviour of the control of visual trajectories. In particular, the exact same model can be examined in the behavioural paradigm of smooth tracking of a moving object. In particular, a typical experiment would be the tracking of a smoothly moving pendulum. It is realistically found that a typical eye movement trace contains discontinuous micro-saccades during smooth pursuit, and discontinuous catch-up saccades. In particular, we expect from the model that the instability will be most present in the periods of stability when kernel deposition is strongest. Further experiments could also find the behaviour of dynamics in phase space $\alpha/\beta/\gamma/t_\theta$ for different neurological conditions. For instance, it is well-known that schizophrenia effects the

Chapter 4: Functional Advantages of Rich Spatiotemporal Searching Dynamics

The power of a well-motivated biologically realistic model of eye movements is in examining functional roles of individual dynamical components studied in the rich task of gathering information. This approach allows an understanding of brain function in the higher layer of network-level neural sampling during visual search. A major motivation for this study is the previous work in the information efficiency of the brain; firstly, that movements are well predicted by greedy informational-maximisation strategies [110, 141, 142], and secondly that this efficiency and function of gaze dynamics during search is a significant indicator of brain health and function [57–63]. This forms a novel connection between the abstract performance of probabilistic sampling and the health and function of visual search performance as a proxy for network-level neurological functioning. Each dynamical component here described has analogy to a feature of abstract sampling algorithms: kernel-driven adaptation will improve ergodicity and can be compared to metadynamics [127, 128, 143], local harmonic oscillations can be compared to the fast mixing and negative autocorrelation in sampling found in Hamiltonian Monte Carlo [122, 144] and second-order Langevin sampling methods [144, 145], and finally Lévy superdiffusion has gathered a growing body of literature as a robust sampling strategy which does not require fine-tuning to explore unknown heavy-tailed or multimodal distributions [118, 123–125].

The first section of this chapter (Section 4.1) presents informational-theoretic methods which have been constructed to analyse efficiency during abstract search tasks. Key to this method is the quantification of information of a belief distribution as the distribution entropy $S_{\pi_t} = -\int_{\mathbf{x}} d\mathbf{x} \ln \pi_t(\mathbf{x})$, and tracking its development through a search task. The second section (Section 4.1) takes these metrics and compares the informational advantages from dynamics driven in parameter space $\alpha/\beta/\gamma/\tau$. The final section (Section 4.2) examines further extensions of these dynamics beyond the case of 2-dimensional visual search to high-dimensional or dynamical target landscapes, where control and exploration are hypothesised be more responsive and robust to shifts in a loss function.

4.1 Functional roles of dynamical patterns

The dynamics studied in this thesis are characterised by three distinct spatiotemporal signatures: gamma-band oscillations, theta-scale rhythmicity, and superdiffusive spatial trajectories. The framework of Chapter 2, which models Equation (2.3) in a search task, unifies these signatures as coupled dynamics that provide flexible functional roles in basic computation. In this section we identify each pattern with a role in sampling: local exploitation in gamma oscillations, temporal organisation in theta rhythmicity, and global exploration through α-superdiffusion. To make these roles precise, we define a set of information-theoretic metrics of search efficiency and, for each metric in turn, describe how each of the three patterns affects it.

The three patterns are controlled by distinct parts of the model:

- **Gamma oscillations** arise from the memory–momentum term p_t (strength β) and the α-Lévy noise dL_t^α within a highly localised potential well. They are analogous to the gamma oscillations found throughout the brain in almost every computational task, and to high-frequency ocular microtremor in gaze.

- **Theta rhythmicity** is the regular, theta-scale timing of transitions between potential wells, tied to the rate of information acquisition through the informative timescale τ (Equation (2.10)). For motion driven by Lévy or Brownian noise alone, transition intervals are instead exponentially distributed according to a Poisson point process, as follows from the Lévy–Itô decomposition theorem (Section 1.2.3).

- **Superdiffusion** is governed by the stability index α and sets the spatial scaling of global exploration. Superdiffusive paths are thought to be optimal for exploring sparse unknown landscapes under the Lévy foraging hypothesis (Section 1.2.2).

4.1.1 Information gain

When considering the efficiency advantages of strategic sampling in a visual search task, the central quantity under study is the information gain, $IG(t)$. To quantify informational search efficiency across an abstract search landscape, we take inspiration from Bayesian inference and sampling methods. In particular, the adaptive dynamics of our sampling model closely resemble well-tempered metadynamics in Markov chain Monte Carlo (MCMC) sampling [143]. Similar techniques have been used to model visual search in discrete time by Bujia et al. [9], where the equivalent target distribution is the likelihood distribution for a search goal, and information is tracked only at the goal position. In Section 2.2 we expanded and unified these approaches by analysing a continuous-time dynamical-information kernel. By tracking the posterior target distribution after kernel deposition at every location in the visual scene, we model global information acquisition across an evolving landscape of belief.

Given a search through a probabilistic landscape for high-probability modes, the information extracted is proportional to the change in entropy of the target distribution. Since we study updates to an effective belief distribution, we use the language and tools of Bayesian inference directly. The initial pre-search belief distribution, $\pi_0(\mathbf{x})$, is defined through the saliency-potential landscape $V_0(\mathbf{x})$ by Boltzmann–Gibbs statistics,

$$\pi_0(\mathbf{x}) = \frac{\exp\{-V_0(\mathbf{x})\}}{\int_{\mathbf{x}'} \exp\{-V_0(\mathbf{x}')\}\, d\mathbf{x}'}. \tag{4.1}$$

As the search progresses, the potential updates in continuous time to encode learned information, through an accumulated kernel over time t,

$$V_t(\mathbf{x}) = V_0(\mathbf{x}) + \int_0^t k(\mathbf{x}_{t'})\, \frac{dt'}{\tau}. \tag{4.2}$$

This adapted potential defines the posterior belief distribution, the updated target likelihood landscape at time t,

$$\pi_t(\mathbf{x}) = \frac{\exp\{-V_t(\mathbf{x})\}}{\int_{\mathbf{x}'} \exp\{-V_t(\mathbf{x}')\}\, d\mathbf{x}'}. \tag{4.3}$$

A natural first step in comparing information gathering through time is the cross-entropy of the posterior to the prior,

$$H(\pi_t, \pi_0) = -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x}. \tag{4.4}$$

However, cross-entropy alone is of limited utility, as it conflates the internal concentration of belief within the posterior (its self-entropy, $H(\pi_t)$) with the net information gained relative to the prior. To decouple these contributions we subtract the posterior entropy to obtain the Kullback–Leibler (KL) divergence $D_{\mathrm{KL}}(\pi_t \parallel \pi_0)$, which gives the formal definition of information gain up to time t:

$$\begin{aligned} IG(t) &= H(\pi_t, \pi_0) - H(\pi_t) \\ &= -\int \pi_t(\mathbf{x}) \log \pi_0(\mathbf{x})\, d\mathbf{x} + \int \pi_t(\mathbf{x}) \log \pi_t(\mathbf{x})\, d\mathbf{x} \\ &= \int \pi_t(\mathbf{x}) \log \left(\frac{\pi_t(\mathbf{x})}{\pi_0(\mathbf{x})} \right) d\mathbf{x} \\ &\equiv D_{\mathrm{KL}}(\pi_t \parallel \pi_0). \end{aligned} \tag{4.5}$$

The KL divergence is non-negative and asymmetric. In the context of an abstract sampling model, non-negativity reflects the fact that any spatial update from π_0 represents accumulated evidence, though not necessarily optimal search efficiency. A weak kernel, a slow informative timescale τ, or deposition in low-likelihood areas alters the landscape only marginally and results in slower growth in IG. The information content of the search trajectory is therefore entirely captured by IG, which quantifies the global information acquired as the observer's belief landscape evolves from π_0 to π_t. The metrics in the following subsections summarise different aspects of the curve $IG(t)$.

Since τ sets the timescale of information gathering, time-dependent metrics such as IV and FPT are trivially improved by decreasing τ. When testing the effect of a dynamical pattern, particularly theta-rhythmic transitions, comparisons must therefore be made at fixed τ or through dimensionless metrics.

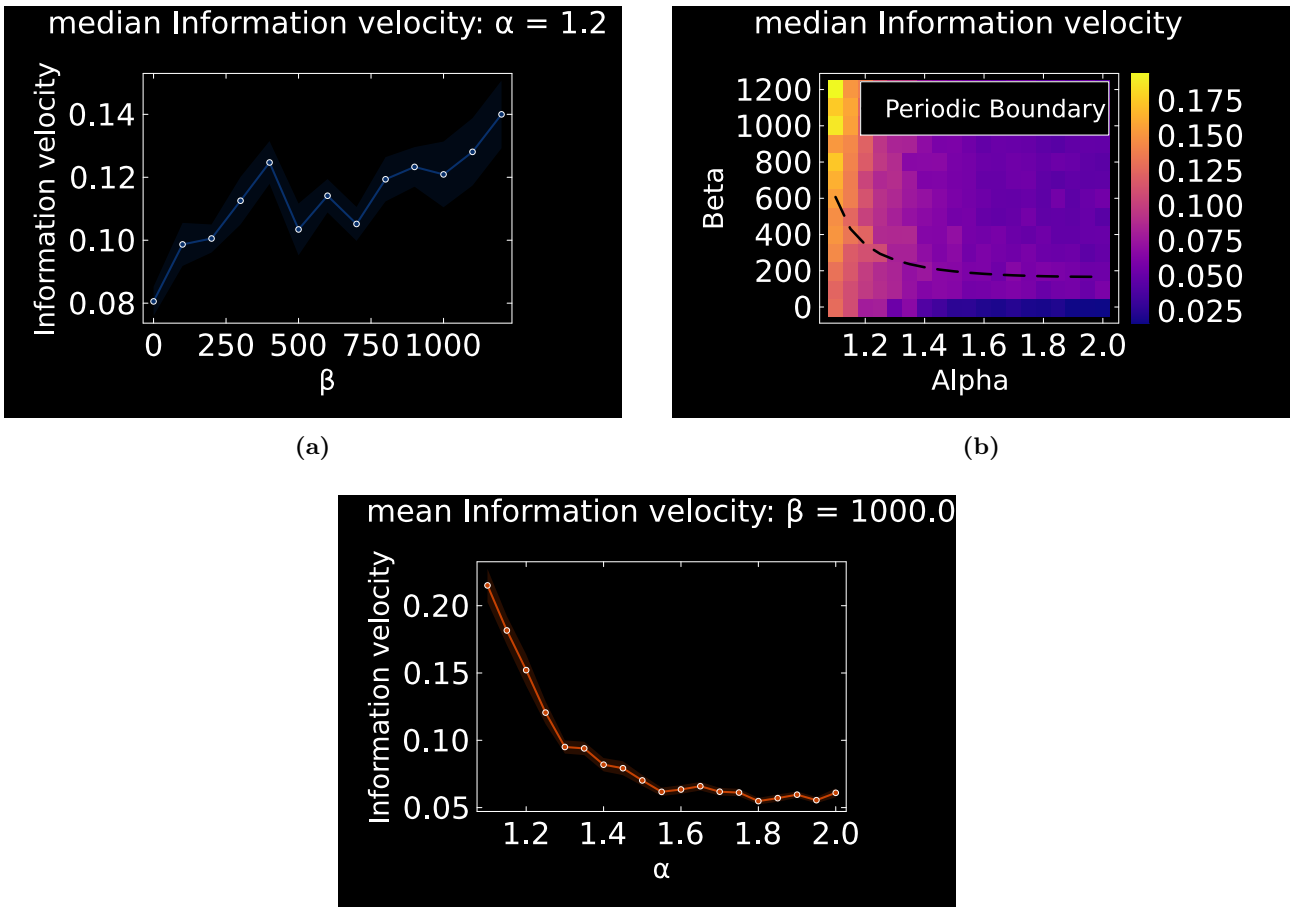
4.1.2 Information velocity

The information velocity, $IV = dIG/dt$, is the first derivative of the information gain. Comparing its averaged value at the same informative timescale τ measures the quality of information gathered by different strategies. A strategy that targets high-likelihood areas, and drives transitions sensitive to decreases in the level of likelihood, will have high average IV_{mean} and IV_{med}.

Gamma oscillations. Rapid local decorrelation means kernel is deposited across the well at many independent locations, so we expect stronger oscillations (larger β) to increase how quickly information is extracted from a high-likelihood region (Figure 4.1a,b).

Theta rhythmicity. Rhythmic transitions end each dwell at a point linked to information acquisition, so time is not wasted in depleted wells. At fixed τ this is expected to raise the average information velocity relative to Poisson-distributed transitions.

Superdiffusion. Long jumps place the walker in new high-likelihood regions where kernel deposition is most informative, which is expected to raise the mean IV relative to purely local diffusion (Figure 4.1c).



(a) **(b)**

(c)

Figure 4.1: Information velocity. (a) Median mean IV against the momentum strength β at $\alpha = 1.2$. (b) Heatmap of median mean IV. (c) Mean IV against the stability index α at $\beta = 1000$.

4.1.3 First passage time

Perhaps the clearest metric of search efficiency is the speed of completing a task. In our system this is the duration of each trial, the first passage time (FPT). We can derive an expected FPT from the theta-scale rate of saccades (Equation (2.10) in Section 2.2) and the number of target points in the visual scene relative to the total. For K targets among N points, the probability of reaching no target after n saccades is the hypergeometric distribution evaluated at zero,

$$p_n(0) = \frac{\binom{K}{0}\binom{N-K}{n}}{\binom{N}{n}} = \frac{(N-K)!\,(N-n)!}{(N-K-n)!\,N!}. \tag{4.6}$$

The expected number of saccades $\mathbb{E}[n]$ is the n for which Equation (4.6) first falls below 0.5, and the expected first passage time is $\mathbb{E}[\mathrm{FPT}] = t_\theta\,\mathbb{E}[n]$. For our standard parameters and experiment, defined in Section 2.1, this gives

$$\begin{aligned}\mathbb{E}[\mathrm{FPT}] &= t_\theta \times \mathbb{E}_{N=256,K=5}[n] \\ &= 0.1422 \times 94 \\ &= 13.36 \text{ s}.\end{aligned} \tag{4.7}$$

This calculation assumes that each transition is independent and drawn from a stationary distribution with mean t_θ across the search task. The true dynamics of the system are therefore expected to yield an averaged FPT that depends on the system parameters, and Equation (4.7) serves as a baseline against which the effect of each pattern is assessed.

Gamma oscillations. The memory–momentum term β keeps the sampler responsive to real-time changes in the saliency landscape as information is received from the scene. This responsiveness is not captured by the baseline, and can shorten the time to find a target.

Theta rhythmicity. The baseline uses the theta rate t_θ directly, so theta rhythmicity sets the mean waiting time per saccade. Regular intervals with the same mean are expected to narrow the spread of first passage times compared with exponentially distributed intervals.

Superdiffusion. A superdiffusive walker has access to the entire scene, and may explore nearby fixation points before migrating to far-displaced areas. This breaks the independence assumed in the baseline, so the mean FPT is expected to depend on α through how efficiently the walker reaches unvisited targets (Figure 4.2).

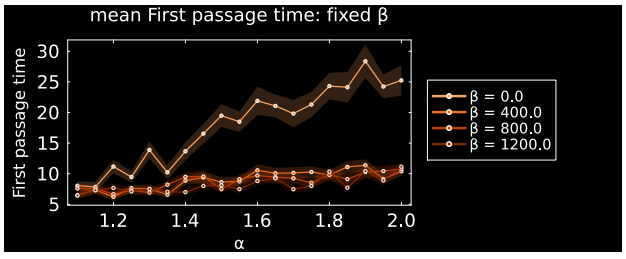


Figure 4.2: First passage time. Mean FPT against the stability index α.

4.1.4 Shape parameter

While the preceding metrics depend explicitly on the search timescale, we can evaluate the fundamental efficiency of a sampling process by making it dimensionless. By normalising each trial so that both time and information content range from 0 to 1, we can examine the shape of the monotonic accumulation curve $IG(t)$ independently of absolute time. Statistically, this is captured by the area under the normalised curve, which can be transformed into an effective power-law exponent for the growth of information gain.

Gamma oscillations. Gamma activity acts at short times, so its influence on the dimensionless curve is expected to appear in how front-loaded information gain is within each visited well.

Theta rhythmicity. Because the shape parameter is dimensionless, it separates the effect of rhythmic transitions from the trivial dependence on τ. Regular transitions avoid long dwells in depleted wells, which should keep the normalised accumulation curve closer to linear rather than saturating early.

Superdiffusion. Trapping in a depleted well causes information gain to plateau early, giving a strongly concave normalised curve. By escaping such wells, superdiffusion is expected to sustain information gain later in the trial and so raise the effective exponent (Figure 4.3).

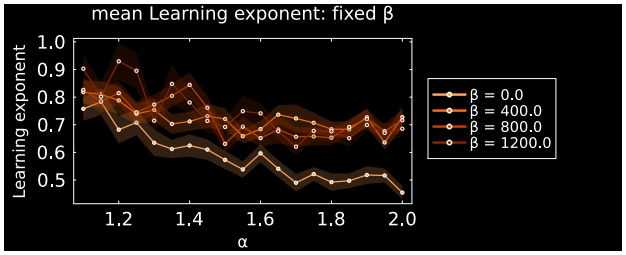


Figure 4.3: Shape parameter. Mean shape exponent of the normalised information-gain curve against the stability index α.

4.1.5 Gini coefficient

To measure the inequality in the information gained, we bin the trial into time windows longer than the gamma oscillation and compute the Gini coefficient of the information gained across bins. A low value indicates that information is acquired evenly through the search, while a high value indicates that it is concentrated in a few periods.

Gamma oscillations. Since the bins are longer than the gamma period, the oscillations themselves are not resolved. The metric instead tests whether stronger local sampling spreads information acquisition more evenly across successive windows (Figure 4.4).

Theta rhythmicity. This is the metric most directly tied to theta organisation. Because rhythmic transitions end each dwell at a point linked to information acquisition, rather than at a random time, we expect a more even distribution of information across time windows, and hence a lower Gini coefficient than for the Poisson case.

Superdiffusion. Long jumps may concentrate information in bursts when a new mode is reached, which would raise the Gini coefficient for small α.

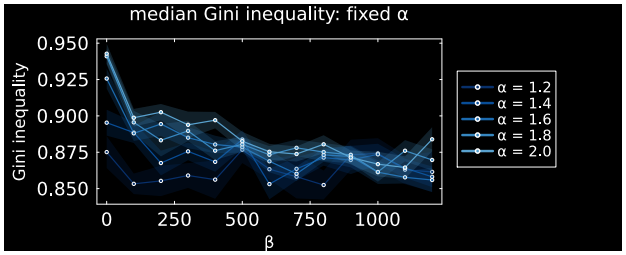


Figure 4.4: Gini coefficient. Median Gini coefficient of binned information gain against the momentum strength β.

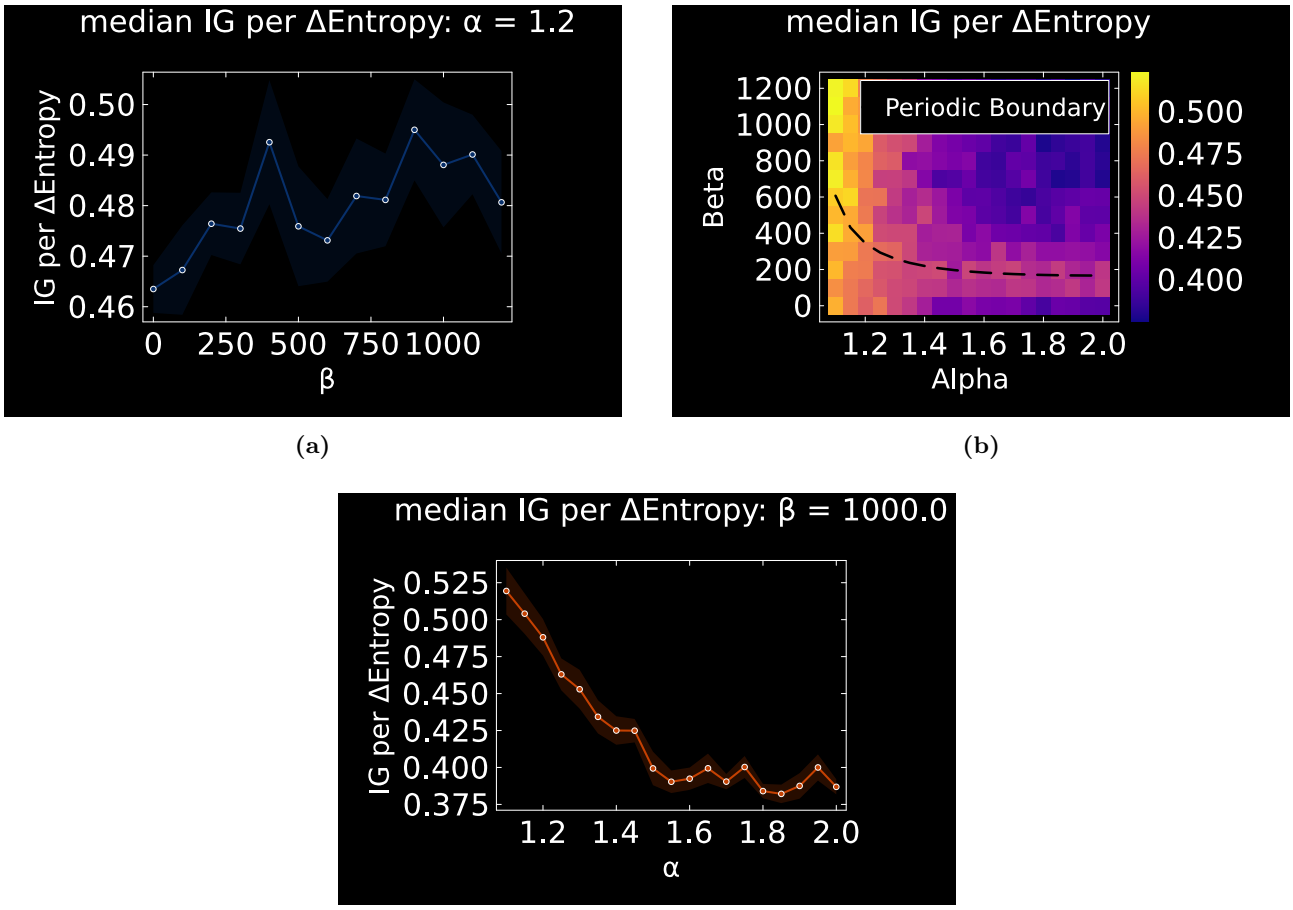
4.1.6 Information–entropy gradient

The information–entropy gradient, $\mathrm{IG_H}$, measures the gradient of information gain with respect to changes in posterior certainty, that is, the efficiency with which a change in belief gains information.

Gamma oscillations. Sampling local gradients directs the deposition of the kernel towards where the posterior is changing, so we expect each unit of belief change to yield more information, raising IG_H (Figure 4.5a,b).

Theta rhythmicity. Theta rhythmicity governs when sampling moves rather than where, so its effect on IG_H is expected to be indirect, through how fully each well is exploited before transition.

Superdiffusion. By visiting unexplored modes, superdiffusion reduces posterior entropy where it is least certain, so we expect it to change the information gained per unit of entropy (Figure 4.5c).



(a) **(b)**

(c)

Figure 4.5: Information–entropy gradient. (a) Median IG_H against the momentum strength β at $\alpha = 1.2$. (b) Heatmap of median IG_H. (c) Median IG_H against the stability index α at $\beta = 1000$.

4.1.7 Summary

Table 4.1 collects the expected effect of each pattern on each metric. Gamma oscillations act at the finest scale to probe local gradients, theta rhythmicity organises when sampling transitions between wells, and superdiffusion sets where the sampler can go. Together they cover complementary aspects of the information-gain curve.

4.2 Further extensions of studied dynamics

4.2.1 Superdiffusive probabilistic sampling

4.2.2 Dynamical goals

Table 4.1: Expected effect of each dynamical pattern on each metric. Entries summarise the proposed mechanisms and are to be confirmed against the results.

Metric	**Gamma oscillations**	**Theta rhythmicity**	**Superdiffusion**
IV	Faster local information extraction	Avoids wasted time in depleted wells	Accesses new high-likelihood regions
FPT	Responds to landscape changes	Sets mean waiting time; narrows spread	Enables global access and escape from local traps
Shape	Front-loads information gain within wells	Keeps accumulation closer to linear at fixed τ	Sustains information gain later in the trial
Gini	Tests evenness across time windows	Promotes more evenly distributed information gain	May produce bursts of information at mode changes
IG_H	Supports gradient-directed deposition	Acts indirectly through well exploitation	Changes information gain through exploration of new modes

Chapter 5: Discussion

In this thesis we have presented the first model to unify the rich multiscale dynamics of visual search. Previous models have focused their object of study on large-scale saccades or small-scale fixations, and motivations for such models are divided in either aiming to present a biologically accurate understanding of visual search, or a black-box description of gaze dynamics as in machine learning. Taking aim to unify these aims and objects of study, we constructed a model from fractional neural sampling (FNS) [91, 126], in which gaze is represented through a second-order stochastic process 2.3 driven by Lévy noise in a saliency landscape that adapts through a dynamical-informative kernel (Chapter 2). We understand these dynamics through three key signatures: global Lévy superdiffusion, theta-rhythmic transitions between regions of interest, and local high-frequency harmonic oscillations within each region. We compared each with the high-resolution gaze data of Näher et al. [41] and with the wider literature, and found that the model captures the characteristic spatiotemporal signatures of saccades and fixations within a single framework (Chapter 3). Finally, we used information-theoretic metrics to show that each pattern plays a distinct functional role in the efficiency of informative search (Chapter 4).

Three ingredients of the model are of general interest in physics. First, the superdiffusion is monoscaling, governed by a single stability index α. We characterised its short-time, noise-driven scaling of fractional absolute moments and the limits of monofractality that such a description can represent, which sets the boundary between what a single exponent can explain and what would require richer multiscaling. Second, the model contains nested oscillations: theta-scale rhythmic transitions between regions, and gamma-scale harmonic oscillations within them, arising from the memory–momentum term and the confining potential. Nested timescales are a recurrent feature of neural dynamics [13, 16], and here they emerge from the same stochastic equation rather than being imposed separately. Third, the dynamical-informative kernel makes the potential history-dependent, so the searcher deposits kernel as it samples and its belief landscape evolves from a prior π_0 to a posterior π_t. The cumulative information gathered is then well defined as a Kullback–Leibler divergence. Together these describe an open, history-dependent, noise-driven system exploring an adaptive landscape, and they give gaze a quantitative language for memory, heavy-tailed fluctuations, and characteristic timescales.

For neuroscience, the model turns gaze into a source of interpretable parameters. The stability index, memory–momentum strength, informative timescale, and kernel strength each have a defined effect on the trajectory, so fitting them to data estimates the dynamics of the underlying attentional control. Several experiments would sharpen this. High-frequency eye tracking without a bite-bar, combined with head acceleration measurements, would resolve gamma-scale fixational dynamics in naturalistic, head-free viewing and separate eye from head contributions. High-quality smooth pursuit experiments, in which the target moves, would probe how the landscape is tracked over time. Dynamical search tasks with controlled likelihood landscapes would constrain the kernel and the theta-rhythmic transitions. Because gaze aggregates the activity of many regions, the fitted parameters may summarise collective properties of the circuits that control and process visual input. A natural candidate is the interaction between the superior colliculus, which integrates sensory, cortical, and goal-related signals into a map that drives saccades [[**CITE**]], and the visual cortex, which supplies the evidence on which that map is built [[**CITE**]]. Within the active vision loop, the model's evolving potential could be compared with such a priority map, and the kernel with processes such as inhibition of return [11, 12], giving a way to study attention in this system quantitatively. Because alterations of eye movements accompany many neurological and psychiatric conditions [[**CITE**]], mechanistic parameters of this kind also offer a route to more accurate biomarkers, and potentially to treatments that target a specific part of the loop.

The same ideas apply to machine learning, in particular to sampling from multimodal and heavy-tailed distributions. Standard Markov chain Monte Carlo methods based on local moves struggle to cross between well-separated modes. The model's superdiffusive jumps address this directly, and

its adaptive kernel resembles well-tempered metadynamics [143], in which deposited bias discourages revisiting explored regions. The link to free-energy minimisation and active inference [[**CITE**], 21] is also close, since both treat search as the reduction of uncertainty about a landscape. Our results emphasise that coupling oscillations across scales is itself important for an abstract sampler: rapid local oscillations explore each mode, while slower rhythmic transitions decide when to move on. Superdiffusive samplers have recently attracted considerable interest [[**CITE**]]. The same principle has influenced more general AI frameworks, notably in work from our group on fractional neural attention (FNA) [**qu˙fractional˙2025**]. FNA models token interactions through Lévy diffusion governed by the fractional Laplacian, so that short- and long-range dependencies arise from a single operator rather than hand-designed local–global patterns. In the reported experiments it matches or exceeds Transformer baselines on text classification, image classification, and machine translation, in some cases with a single layer, and its attention networks show larger spectral gaps and shorter path lengths, consistent with faster information mixing. That the same mechanism describes both gaze and attention in artificial networks supports the view that multiscale, nonlocal sampling is a general computational strategy.

In conclusion, this thesis has shown that the rich and apparently disparate dynamics of human gaze can be described by a single physical model with a small set of interpretable ingredients: nonlocal Lévy jumps, nested oscillations, and an adaptive, information-gathering landscape. Gaze is both easily measured and strongly linked to neural decision-making, so understanding its dynamics gives access to basic computation in the brain, namely how it samples, remembers, and decides under uncertainty. The model offers a quantitative framework for this, makes testable predictions for new experiments, and links vision to wider questions in statistical physics and artificial intelligence. Much remains to be done, in particular multiscaling beyond a single exponent, direct tests in neural recordings, and clinical application, but we hope the framework provides a firm basis for this work.

Appendix A: Prior models of visual search

Model	Core domain	Primary aim	Key features
Ideal Bayesian searcher (IBS) Najemnik and Geisler [112], see also [146, 147]	Information- and decision theory	Establish an optimally informative process for saccade localisation based on biologically realistic information foraging of the eye.	Exact Bayesian belief updating, incorporating biologically realistic retinal resolution gradients, spatial noise, and global entropy reduction. Considers discrete saccades rather than continuous time.
Least action / active inference Zanca and Gori [21]	Machine learning	Generating scanpaths (trajectories) through physics-inspired variational informational free-energy minimisation.	Generative deterministic model of scanpaths. Sophisticated account of natural image saliency. Does not consider fixational dynamics, does not aim to be biologically realistic.
Self-avoiding random walk Engbert et al. [7]	Statistical physics and oculomotor control	Develop an accurate account of low-frequency ocular drifts during free-viewing fixations.	Lattice-based stochastic motion with self-avoidance constraints; 2D quadratic potential. Does not capture a high-frequency tremor or saccades.
DeepGaze Abdelrahman, Al-Tawil, and Al-Hamadi [30], see also [10, 34, 37, 38]	Deep learning and computer vision	Construct a neural network to predict human scanpaths on natural images with high empirical accuracy.	Modular deep neural network trained on eye-tracking data; models internal state, memory, and information. Not designed as a biologically realistic model; considers discrete saccade locations.
Itti–Koch–Niebur model Itti, Koch, and Niebur [148]	Biologically inspired vision	Simulate bottom-up, stimulus-driven visual attention based on early primate visual pathways.	Multi-scale feature extraction (colour, intensity, orientation); winner-take-all selection with inhibition of return (IOR).
Feature-integration theory Gayet and Aydemir [149]	Cognitive psychology	Explain experimental differences in performance during sparse- and dense-search tasks.	Pre-attentive direction of attention based on saliency theory. Biologically motivated, but not generative or explanatory for eye movement measurement.
Guided search Wolfe, Cave, and Franzel [150] and Wolfe [151]	Cognitive psychology	Account for the continuum of multi-scale search efficiency by combining bottom-up and top-down guidance of attention.	Parallel feature maps weighted by stimulus salience and target template to form an activation map; serial deployment of attention in decreasing activation order.

Table A.1: Review of a selection of influential models of visual search and gaze dynamics. Comparison across core theoretical domains, primary objectives, and defining operational features. Models span a spectrum from biologically grounded frameworks prioritising physiological mechanisms (e.g., Feature-Integration Theory, Itti–Koch–Niebur) to normative and computational architectures optimised for empirical trajectory prediction on complex natural scenes (e.g., Ideal Bayesian Searcher, DeepGaze). Important features which share analogy in the model presented in this thesis include: an information kernel (in discrete time) from the approach in Najemnik and Geisler [112], the second-order (noiseless) dynamical system coupled to a rich account of saliency gradients presented in Zanca and Gori [21], and the description of the rich non-tremor fixational random motion in the presented in Engbert et al. [7].

Bibliography

[1] B.A. Wandell, S.O. Dumoulin, and A.A. Brewer. "Visual Cortex in Humans". en. In: *Encyclopedia of Neuroscience*. Elsevier, 2009, pp. 251–257. ISBN: 978-0-08-045046-9. DOI: 10.1016/B978-008045046-9.00241-2.

[2] Timothy J. Andrews, Scott D. Halpern, and Dale Purves. "Correlated Size Variations in Human Visual Cortex, Lateral Geniculate Nucleus, and Optic Tract". en. In: *Journal of Neuroscience* 17.8 (1997), pp. 2859–2868. ISSN: 0270-6474, 1529-2401. DOI: 10.1523/JNEUROSCI.17-08-02859.1997.

[3] Zhao-Ping Li. "Vision as looking and seeing through a bottleneck". In: *Current Opinion in Neurobiology* 98 (2026), p. 103211. ISSN: 0959-4388. DOI: 10.1016/j.conb.2026.103211.

[4] D. H. Kelly. "Information capacity of a single retinal channel". In: *IRE Transactions on Information Theory* 8.3 (1962), pp. 221–226. DOI: 10.1109/TIT.1962.1057716.

[5] G. C. Sziklai. "Some studies in the speed of visual perception". In: *Electrical Engineering* 76.2 (1956), pp. 150–152. DOI: 10.1109/TIT.1956.1056814.

[6] Casper J. Erkelens. "Coordination of smooth pursuit and saccades". en. In: *Vision Research* 46.1-2 (2006), pp. 163–170. ISSN: 00426989. DOI: 10.1016/j.visres.2005.06.027.

[7] Ralf Engbert et al. "An integrated model of fixational eye movements and microsaccades". en. In: *Proceedings of the National Academy of Sciences of the United States of America* 108.39 (2011). ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.1102730108.

[8] Neil D. B. Bruce and John K. Tsotsos. "Saliency, attention, and visual search: An information theoretic approach". In: *Journal of Vision* 9.3 (Mar. 2009), pp. 5–5. ISSN: 1534-7362. DOI: 10.1167/9.3.5.

[9] Gaston Bujia et al. "Modeling Human Visual Search in Natural Scenes: A Combined Bayesian Searcher and Saliency Map Approach". en. In: *Frontiers in Systems Neuroscience* 16 (2022). ISSN: 1662-5137. DOI: 10.3389/fnsys.2022.882315.

[10] Xun Huang et al. "SALICON: Reducing the Semantic Gap in Saliency Prediction by Adapting Deep Neural Networks". en. In: *IEEE International Conference on Computer Vision*. Santiago, Chile: IEEE, 2015, pp. 262–270. ISBN: 978-1-4673-8391-2. DOI: 10.1109/ICCV.2015.38.

[11] Casimir J.H. Ludwig et al. "The mechanism underlying inhibition of saccadic return". en. In: *Cognitive Psychology* 59.2 (2009), pp. 180–202. ISSN: 00100285. DOI: 10.1016/j.cogpsych.2009.04.002.

[12] Ralph S. Redden, W. Joseph MacInnes, and Raymond M. Klein. "Inhibition of return: An information processing theory of its natures and significance". en. In: *Cortex; a journal devoted to the study of the nervous system and behavior* 135 (2020), pp. 30–48. ISSN: 00109452. DOI: 10.31234/osf.io/s29f5.

[13] Enda Tan et al. "Theta activity and cognitive functioning: Integrating evidence from resting-state and task-related developmental electroencephalography (EEG) research". en. In: *Developmental Cognitive Neuroscience* 67 (2024). ISSN: 18789293. DOI: 10.1016/j.dcn.2024.101404.

[14] Tomoya Kawashima, Masamichi J Hayashi, and Kaoru Amano. *Attentional rhythmic blink: Theta/Alpha balance in neural oscillations determines the rhythmicity in visual sampling*. en. 2022. DOI: 10.1101/2022.04.15.488436.

[15] Humza N. Zubair et al. "Eye movements reflect memory-related theta activity in the human brain". en. In: *PLOS Biology* 24.3 (2026). Ed. by Raphael Kaplan, e3003695. ISSN: 1545-7885. DOI: 10.1371/journal.pbio.3003695.

[16] Georgios Spyropoulos, Conrado Arturo Bosman, and Pascal Fries. "A theta rhythm in macaque visual cortex and its attentional modulation". en. In: *Proceedings of the National Academy of Sciences of the United States of America* 115.24 (2018). ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.1719433115.

[17] Hee-kyoung Ko, D. Max Snodderly, and Martina Poletti. "Eye movements between saccades: Measuring ocular drift and tremor". en. In: *Vision Research* 122 (2016), pp. 93–104. ISSN: 00426989. DOI: 10.1016/j.visres.2016.03.006.

[18] Lisa Graham et al. "Ocular microtremor: a structured review". en. In: *Experimental Brain Research* 241.9 (2023), pp. 2191–2203. ISSN: 0014-4819, 1432-1106. DOI: 10.1007/s00221-023-06691-w.

[19] Alexander Spauschus et al. "The origin of ocular microtremor in man". en. In: *Experimental Brain Research* 126.4 (1999), pp. 556–562. ISSN: 00144819. DOI: 10.1007/s002210050764.

[20] Michele Rucci and Martina Poletti. "Control and Functions of Fixational Eye Movements". en. In: *Annual Review of Vision Science* 1.1 (2015), pp. 499–518. ISSN: 2374-4642, 2374-4650. DOI: 10.1146/annurev-vision-082114-035742.

[21] Dario Zanca and Marco Gori. "Variational Laws of Visual Attention for Dynamic Scenes". In: *Neural Information Processing Systems*. Ed. by I. Guyon et al. Vol. 30. Curran Associates, Inc., 2017.

[22] Piotr Błażejczyk and Marcin Magdziarz. "Stochastic modeling of Lévy-like human eye movements". en. In: *Chaos: An Interdisciplinary Journal of Nonlinear Science* 31.4 (2021). ISSN: 1054-1500, 1089-7682. DOI: 10.1063/5.0036491.

[23] Pedro Lencastre et al. "Dynamical law behind eye movements: Distinguishing between Lévy and intermittent strategies". en. In: *Physical Review Research* 7.4 (2025). ISSN: 2643-1564. DOI: 10.1103/2bzm-t9k1.

[24] R Vasudevan, A V Phatak, and J D Smith. "A stochastic model for eye movements during fixation on a stationary target". In: *Kybernetik* 11.1 (1972), pp. 264–270. DOI: 10.1007/BF00267762.

[25] Xutao Kuang et al. "Temporal Encoding of Spatial Information during Active Visual Fixation". en. In: *Current Biology* 22.6 (2012), pp. 510–514. ISSN: 09609822. DOI: 10.1016/j.cub.2012.01.050.

[26] Susana Martinez-Conde, Stephen L. Macknik, and David H. Hubel. "The role of fixational eye movements in visual perception". en. In: *Nature Reviews Neuroscience* 5.3 (2004), pp. 229–240. ISSN: 1471-003X, 1471-0048. DOI: 10.1038/nrn1348.

[27] Jorge Otero-Millan, Stephen L. Macknik, and Susana Martinez-Conde. "Fixational eye movements and binocular vision". en. In: *Frontiers in Integrative Neuroscience* 8 (2014). ISSN: 1662-5145. DOI: 10.3389/fnint.2014.00052.

[28] Shailendra Bhandari et al. "Modeling eye gaze velocity trajectories using GANs with spectral loss for enhanced fidelity". en. In: *Scientific Reports* 15.1 (2024). ISSN: 2045-2322. DOI: 10.1038/s41598-025-05286-5.

[29] Shaoxuan Wu et al. "GiTNet: A graph-based trajectory-informed network for gaze-supervised medical image segmentation". en. In: *Medical Image Anal.* 112 (2026), p. 104112. ISSN: 13618415. DOI: 10.1016/j.media.2026.104112.

[30] Ahmed A. Abdelrahman, Basheer Al-Tawil, and Ayoub Al-Hamadi. "Deep Learning-Based Gaze Estimation: A Review". en. In: *Robotics* 15.4 (2026), p. 69. ISSN: 2218-6581. DOI: 10.3390/robotics15040069.

[31] Gianni Bremer, Niklas Stein, and Markus Lappe. "Predicting Future Position From Natural Walking and Eye Movements with Machine Learning". en. In: *International Conference on Artificial Intelligence and Virtual Reality*. Taichung, Taiwan: IEEE, 2021, pp. 19–28. ISBN: 978-1-6654-3225-2. DOI: 10.1109/AIVR52153.2021.00013.

[32] Süleyman Özdel et al. "Gaze-Guided Graph Neural Network for Action Anticipation Conditioned on Intention". en. In: *Eye Tracking Research & Application*. Glasgow United Kingdom: ACM, 2024, pp. 1–9. ISBN: 979-8-4007-0607-3. DOI: 10.1145/3649902.3653340.

[33] Andronicus A. Akinyelu and Pieter Blignaut. "Convolutional Neural Network-Based Technique for Gaze Estimation on Mobile Devices". en. In: *Frontiers in Artificial Intelligence* 4 (2022). ISSN: 2624-8212. DOI: 10.3389/frai.2021.796825.

[34] Matthias Kümmerer, Harneet Singh Khanuja, and Matthias Bethge. *Modeling Saliency Dataset Bias*. en. 2025. DOI: 10.1109/ICCV51701.2025.02050.

[35] Marcella Cornia et al. "A deep multi-level network for saliency prediction". en. In: *International Conference on Pattern Recognition*. Cancun: IEEE, 2016, pp. 3488–3493. ISBN: 978-1-5090-4847-2. DOI: 10.1109/ICPR.2016.7900174.

[36] Marcella Cornia et al. "Predicting Human Eye Fixations via an LSTM-Based Saliency Attentive Model". en. In: *IEEE Transactions on Image Processing* 27.10 (2016), pp. 5142–5154. ISSN: 1057-7149, 1941-0042. DOI: 10.1109/TIP.2018.2851672.

[37] Matthias Kümmerer and M. Bethge. "DeepGaze II: A Big Step Towards Explaining All Information in Image-Based Saliency". In: *Journal of Vision* 16.12 (2016), p. 330. DOI: 10.1167/16.12.330.

[38] Matthias Kümmerer, Matthias Bethge, and Thomas S. A. Wallis. "DeepGaze III: Modeling free-viewing human scanpaths with deep learning". In: *Journal of Vision* 22.5 (2022), p. 7. ISSN: 1534-7362. DOI: 10.1167/jov.22.5.7.

[39] Federico D'Agostino et al. "What Moves the Eyes: Doubling Mechanistic Model Performance Using Deep Networks to Discover and Test Cognitive Hypotheses". In: *Neural Information Processing Systems*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 63059–63088. DOI: 10.52202/085713-1895.

[40] Matthias Kümmerer et al. "Understanding Low- and High-Level Contributions to Fixation Prediction". In: Venice: IEEE, 2017, pp. 4799–4808. ISBN: 978-1-5386-1032-9. DOI: 10.1109/ICCV.2017.513.

[41] Tim Näher et al. "Primate Saccade Rhythmicity". In: *bioRxiv* (2026). DOI: 10.1101/2023.09.27.559710.

[42] Karl Friston. "Does predictive coding have a future?" en. In: *Nature Neuroscience* 21.8 (2018), pp. 1019–1021. ISSN: 1097-6256, 1546-1726. DOI: 10.1038/s41593-018-0200-7.

[43] John E. Lisman and Ole Jensen. "The Theta-Gamma Neural Code". en. In: *Neuron* 77.6 (2013), pp. 1002–1016. ISSN: 08966273. DOI: 10.1016/j.neuron.2013.03.007.

[44] Douglas McLelland and Rufin VanRullen. "Theta-Gamma Coding Meets Communication-through-Coherence: Neuronal Oscillatory Multiplexing Theories Reconciled". en. In: *PLoS Comput. Biol.* 12.10 (2016). Ed. by Boris S. Gutkin, e1005162. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1005162.

[45] Mauro Ursino and Gabriele Pirazzini. "Theta–gamma coupling as a ubiquitous brain mechanism: implications for memory, attention, dreaming, imagination, and consciousness". en. In: *Current Opinion in Behavioral Sciences* 59 (2024), p. 101433. ISSN: 23521546. DOI: 10.1016/j.cobeha.2024.101433.

[46] J. Fan et al. "The Relation of Brain Oscillations to Attentional Networks". en. In: *Journal of Neuroscience* 27.23 (2007), pp. 6197–6206. ISSN: 0270-6474, 1529-2401. DOI: 10.1523/JNEUROSCI.1833-07.2007.

[47] Wolfgang Klimesch. "The frequency architecture of brain and brain body oscillations: an analysis". en. In: *European Journal of Neuroscience* 48.7 (2018), pp. 2431–2453. ISSN: 0953-816X, 1460-9568. DOI: 10.1111/ejn.14192.

[48] Wenjing Wang, Guoao Liu, and Yang Xi. "Attention Modulates the Neural Oscillation of Theta Frequency in Audiovisual Integration". en. In: *HP3C*. Jilin China: ACM, 2022, pp. 163–167. ISBN: 978-1-4503-9629-5. DOI: 10.1145/3546000.3546025.

[49] Xi Yang et al. "The neural oscillations in delta- and theta-bands contribute to divided attention in audiovisual integration". en. In: *Perception* 53.1 (2023), pp. 44–60. ISSN: 0301-0066, 1468-4233. DOI: 10.1177/03010066231208539.

[50] Ce Mo et al. "Competing rhythmic neural representations of orientations during concurrent attention to multiple orientation features". en. In: *Nature Communications* 10.1 (2019). DOI: 10.1038/s41467-019-13282-3.

[51] R. T. Canolty et al. "High Gamma Power Is Phase-Locked to Theta Oscillations in Human Neocortex". en. In: *Science* 313.5793 (2006), pp. 1626–1628. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.1128115.

[52] Brendan Harris and Pulin Gong. "Nested spatiotemporal theta-gamma waves organize hierarchical processing across the mouse visual cortex". In: *Nature Communications* 17.1 (2026). DOI: 10.1038/s41467-026-68893-4.

[53] Heather J. Brooks et al. "Theta–gamma coupling and ordering information: a stable brain–behavior relationship across cognitive tasks and clinical conditions". en. In: *Neuropsychopharmacology* 45.12 (2020), pp. 2038–2047. ISSN: 0893-133X, 1740-634X. DOI: 10.1038/s41386-020-0759-z.

[54] Adriano B. L. Tort et al. "Theta–gamma coupling increases during the learning of item–context associations". en. In: *Proceedings of the National Academy of Sciences of the United States of America* 106.49 (2009), pp. 20942–20947. ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.0911331106.

[55] H. S. Seung. "How the brain keeps the eyes still". en. In: *Proceedings of the National Academy of Sciences of the United States of America* 93.23 (1996), pp. 13339–13344. ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.93.23.13339.

[56] J. Benitez. "Eye-tracking and optokinetic tests: Diagnostic significance in peripheral and central vestibular disorders". en. In: *The Laryngoscope* 80.6 (1970). ISSN: 0023-852X, 1531-4995. DOI: 10.1288/00005537-197006000-00001.

[57] Christos Papanikolaou et al. "Lévy Flight Model of Gaze Trajectories to Assist in ADHD Diagnoses". en. In: *Entropy* 26.5 (2024), p. 392. ISSN: 1099-4300. DOI: 10.3390/e26050392.

[58] Simona Caldani et al. "Saccadic eye movements as markers of schizophrenia spectrum: Exploration in at-risk mental states". en. In: *Schizophrenia Research* 181 (2017), pp. 30–37. ISSN: 09209964. DOI: 10.1016/j.schres.2016.09.003.

[59] Kentaro Morita et al. "Eye movement as a biomarker of schizophrenia: Using an integrated eye movement score". en. In: *Psychiatry and Clinical Neurosciences* 71.2 (2017), pp. 104–114. ISSN: 1323-1316, 1440-1819. DOI: 10.1111/pcn.12460.

[60] Jiahui Zhu et al. "Diagnosis of schizophrenia by integrated saccade scores and associations with psychiatric symptoms, and functioning". In: *Medicine* 103.41 (2024), e39935. ISSN: 1536-5964. DOI: 10.1097/MD.0000000000039935.

[61] Axel Lindner et al. "Disorders of Agency in Schizophrenia Correlate with an Inability to Compensate for the Sensory Consequences of Actions". en. In: *Current Biology* 15.12 (2005), pp. 1119–1124. ISSN: 09609822. DOI: 10.1016/j.cub.2005.05.049.

[62] Juliana Bittencourt et al. "Saccadic eye movement applications for psychiatric disorders". en. In: *Neuropsychiatric Disease and Treatment* (2013), p. 1393. ISSN: 1176-6328. DOI: 10.2147/NDT.S45931.

[63] Yong-Lin Wang et al. "Saccadic eye movements in neurological disease: cognitive mechanisms and clinical applications". en. In: *Journal of Neurology* 272.8 (2025). ISSN: 0340-5354, 1432-1459. DOI: 10.1007/s00415-025-13275-x.

[64] F. van Ede. "Looking into working memory through micro eye movements". en. In: *Trends in Cognitive Sciences* (2026). ISSN: 13646613. DOI: 10.1016/j.tics.2026.06.001.

[65] Uwe Hassler. *Stochastic Processes and Calculus: An Elementary Introduction with Applications*. en. Springer Texts in Business and Economics. Cham: Springer International Publishing, 2015. ISBN: 978-3-319-23427-4. DOI: 10.1007/978-3-319-23428-1.

[66] Aashrit Cunchala. *A Basic Overview of Various Stochastic Approaches to Financial Modeling With Examples*. 2024.

[67] Sergei Izvekov. "Erratum: Mori-Zwanzig projection operator formalism: Generalized Langevin equation dynamics of a classical system perturbed by an external generalized potential and far from equilibrium [Phys. Rev. E 111, 034130 (2025)]". In: *Physical Review E* 113.2 (2 Feb. 2026), p. 029904. DOI: 10.1103/18k5-pwl1.

[68] G. La Camera. "The mean field approach for populations of spiking neurons". en. In: *Advances in Experimental Medicine and Biology*. Advances in experimental medicine and biology 1359 (2021), pp. 125–157. DOI: 10.1007/978-3-030-89439-9_6.

[69] S. Chandrasekhar. "Dynamical Friction. I. General Considerations: The Coefficient of Dynamical Friction". In: *The Astrophysical Journal* 97 (1943), p. 255. DOI: 10.1086/144517.

[70] David Merritt, Peter Berczik, and Frederik Laun. "Brownian Motion of Black Holes in Dense Nuclei". In: *arXiv* 133.2 (2007), pp. 553–563. DOI: 10.1086/510294.

[71] A. Einstein. "Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen". In: *Annalen der Physik* 322.8 (1905), pp. 549–560. DOI: 10.1002/andp.19053220806.

[72] Joseph Klafter and Igor M Sokolov. "Anomalous diffusion spreads its wings". en. In: *Physics World* 18.8 (2005), pp. 29–32. ISSN: 0953-8585, 2058-7058. DOI: 10.1088/2058-7058/18/8/33.

[73] Ralf Metzler et al. "Anomalous diffusion models and their properties: non-stationarity, non-ergodicity, and ageing at the centenary of single particle tracking". en. In: *Physical Chemistry, Chemical Physics - PCCP* 16.44 (2014), pp. 24128–24164. ISSN: 1463-9076, 1463-9084. DOI: 10.1039/C4CP03465A.

[74] James W Kirchner, Xiahong Feng, and Colin Neal. "Fractal stream chemistry and its implications for contaminant transport in catchments". In: *Nature* 403.6769 (2000), pp. 524–527. DOI: 10.1038/35000537.

[75] S C Kou and X Sunney Xie. "Generalized Langevin equation with fractional Gaussian noise: subdiffusion within a single protein molecule". In: *Physical Review Letters* 93.18 (2004), p. 180603. DOI: 10.1103/PHYSREVLETT.93.180603.

[76] G. Viswanathan et al. "Optimizing the success of random searches". en. In: *Nature* 401.6756 (1999), pp. 911–914. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/44831.

[77] Graham H. Pyke, Andrew G. Kepert, and William J. Chivers. "Lévy foraging hypothesis is a model artefact". In: *Ecological Modelling* 519 (2026), p. 111647. ISSN: 0304-3800. DOI: 10.1016/j.ecolmodel.2026.111647.

[78] Andy Reynolds. "Liberating Lévy walk research from the shackles of optimal foraging". en. In: *Physics of Life Reviews* 14 (2015), pp. 59–83. ISSN: 15710645. DOI: 10.1016/j.plrev.2015.03.002.

[79] Simon Benhamou and Julien Collet. "Ultimate failure of the Lévy Foraging Hypothesis: Two-scale searching strategies outperform scale-free ones even when prey are scarce and cryptic". en. In: *Journal of Theoretical Biology* 387 (2015), pp. 221–227. ISSN: 00225193. DOI: 10.1016/j.jtbi.2015.09.034.

[80] Graham H. Pyke. "Understanding movements of organisms: it's time to abandon the Lévy foraging hypothesis". en. In: *Methods in Ecology and Evolution* 6.1 (2015). Ed. by Luca Giuggioli, pp. 1–16. ISSN: 2041-210X, 2041-210X. DOI: 10.1111/2041-210X.12298.

[81] Nicolas E. Humphries and David W. Sims. "Optimal foraging strategies: Lévy walks balance searching and patch exploitation under a very broad range of conditions". en. In: *Journal of Theoretical Biology* 358 (2014), pp. 179–193. ISSN: 00225193. DOI: 10.1016/j.jtbi.2014.05.032.

[82] Winston Campeau, Andrew M. Simons, and Brett Stevens. "The evolutionary maintenance of Lévy flight foraging". en. In: *bioRxiv* 18.1 (2021). Ed. by Marcos Gomes Eleuterio Da Luz, e1009490. ISSN: 1553-7358. DOI: 10.1101/2021.09.29.462284.

[83] David W Sims et al. "Optimal searching behaviour generated intrinsically by the central pattern generator for locomotion". en. In: *eLife* 8 (2019). ISSN: 2050-084X. DOI: 10.7554/eLife.50316.

[84] A. Clementi et al. "Search via Parallel Lévy Walks on Z2". en. In: *ACM SIGACT-SIGOPS Symposium on Principles of Distributed Computing*. Virtual Event Italy: ACM, 2020, pp. 81–91. ISBN: 978-1-4503-8548-0. DOI: 10.1145/3465084.3467921.

[85] Andy M. Reynolds and Mark A. Frye. "Free-Flight Odor Tracking in Drosophila Is Consistent with an Optimal Intermittent Scale-Free Search". en. In: *PLoS ONE* 2.4 (2007). Ed. by Sean Rands, e354. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0000354.

[86] Stefano Focardi and Jacopo G. Cecere. "The Lévy flight foraging hypothesis in a pelagic seabird". en. In: *Journal of Animal Ecology* 83.2 (2014). Ed. by Graeme Hays, pp. 353–364. ISSN: 0021-8790, 1365-2656. DOI: 10.1111/1365-2656.12147.

[87] Nicolas E. Humphries, Henri Weimerskirch, and David W. Sims. "A new approach for objective identification of turns and steps in organism movement data relevant to random walk modelling". en. In: *Methods in Ecology and Evolution* 4.10 (2013). Ed. by Robert Freckleton, pp. 930–938. ISSN: 2041-210X, 2041-210X. DOI: 10.1111/2041-210X.12096.

[88] David W. Sims et al. "Scaling laws of marine predator search behaviour". en. In: *Nature* 451.7182 (2008), pp. 1098–1102. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/nature06518.

[89] Nicolas E. Humphries et al. "Environmental context explains Lévy and Brownian movement patterns of marine predators". en. In: *Nature* 465.7301 (2010), pp. 1066–1069. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/nature09116.

[90] Arild O Gautestad and Atle Mysterud. "The Lévy flight foraging hypothesis: forgetting about memory may lead to false verification of Brownian motion". en. In: *Movement Ecology* 1.1 (2013). ISSN: 2051-3933. DOI: 10.1186/2051-3933-1-9.

[91] Yang Qi and Pulin Gong. "Fractional neural sampling as a theory of spatiotemporal probabilistic computations in neural circuits". en. In: *Nature Communications* 13.1 (2022). ISSN: 2041-1723. DOI: 10.1038/s41467-022-32279-z.

[92] Ohad Vilk et al. "Strong Anomalous Diffusion for Free-Ranging Birds". en. In: *PRX Life* 3.3 (2024). ISSN: 2835-8279. DOI: 10.1103/yqpg-tn5y.

[93] Mark van den Bosch. "Notes on stochastic integration theory with respect to càdlàg semimartingales and a brief introduction to Lévy processes". In: *arXiv (Cornell University)* (Dec. 2024), pp. 1–70. DOI: 10.48550/arXiv.2501.00140.

[94] Philippe Soulier. "The tail process and tail measure of continuous time regularly varying stochastic processes". In: *Extremes* 25.1 (2020), pp. 107–173. DOI: 10.1007/S10687-021-00417-3.

[95] Vivek S. Borkar. "Foundations of Continuous-Time Processes". In: *Universitext*. New York, NY: Springer New York, 1995, pp. 109–129. ISBN: 978-1-4612-0791-7. DOI: 10.1007/978-1-4612-0791-7_6.

[96] Andreas E. Kyprianou. "The Lévy–Itô Decomposition and Path Structure". In: *Universitext*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2014, pp. 35–69. ISBN: 978-3-642-37632-0. DOI: 10.1007/978-3-642-37632-0_2.

[97] Jean Delacour, J.C Lévy, and D Mercier. "A 'Neural Sampling Theory (NST)' of learning and memory mechanisms". en. In: *Biosystems* 44.3 (1997), pp. 231–244. ISSN: 03032647. DOI: 10.1016/S0303-2647(97)00055-5.

[98] Kyle Cranmer et al. "Advances in machine-learning-based sampling motivated by lattice quantum chromodynamics". en. In: *Nature Reviews Physics* 5.9 (2023), pp. 526–535. ISSN: 2522-5820. DOI: 10.1038/s42254-023-00616-w.

[99] Rajesh P. N. Rao and Dana H. Ballard. "Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects". en. In: *Nature Neuroscience* 2.1 (1999), pp. 79–87. ISSN: 1097-6256, 1546-1726. DOI: 10.1038/4580.

[100] Christof Koch and Tomaso Poggio. "Predicting the visual world: silence is golden". en. In: *Nature Neuroscience* 2.1 (1999), pp. 9–10. ISSN: 1097-6256, 1546-1726. DOI: 10.1038/4511.

[101] Richard D Lange et al. "Bayesian Encoding and Decoding as Distinct Perspectives on Neural Coding". en. In: *bioRxiv* (2020). DOI: 10.1038/s41593-023-01458-6.

[102] Gergő Orbán et al. "Neural Variability and Sampling-Based Probabilistic Representations in the Visual Cortex". en. In: *Neuron* 92.2 (2016), pp. 530–543. ISSN: 08966273. DOI: 10.1016/j.neuron.2016.09.038.

[103] P. Hoyer and Aapo Hyvärinen. "Interpreting Neural Response Variability as Monte Carlo Sampling of the Posterior". In: *Neural Information Processing Systems*. Ed. by S. Becker, S. Thrun, and K. Obermayer. Vol. 15. MIT Press, 2002.

[104] Balazs B Ujfalussy and Gergő Orbán. "Sampling motion trajectories during hippocampal theta sequences". en. In: *bioRxiv* 11 (2021). ISSN: 2050-084X. DOI: 10.1101/2021.12.14.472575.

[105] Yu Terada and Taro Toyoizumi. "Chaotic neural dynamics facilitate probabilistic computations through sampling". en. In: *bioRxiv* 121.18 (2023). ISSN: 0027-8424, 1091-6490. DOI: 10.1073/pnas.2312992121.

[106] Xingsi Dong and Si Wu. "Neural Sampling in Hierarchical Exponential-family Energy-based Models". In: *Advances in Neural Information Processing Systems 36* (2023), pp. 78593–78606. DOI: 10.52202/075280-3437.

[107] V. Koren et al. "Efficient coding in biophysically realistic excitatory-inhibitory spiking networks". en. In: *eLife* 13 (2025). ISSN: 2050-084X. DOI: 10.7554/eLife.99545.

[108] Joseph A. Kilgore et al. "Biologically-informed excitatory and inhibitory ratio for robust spiking neural network training". en. In: *Scientific Reports* 15.1 (2025). ISSN: 2045-2322. DOI: 10.1038/s41598-025-03408-7.

[109] H. Francis Song, Guangyu R. Yang, and Xiao-Jing Wang. "Training Excitatory-Inhibitory Recurrent Neural Networks for Cognitive Tasks: A Simple and Flexible Framework". en. In: *PLOS Computational Biology* 12.2 (2016). Ed. by Olaf Sporns, e1004792. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1004792.

[110] Laura Walker Renninger et al. "An information maximization model of eye movements". en. In: *Neural Information Processing Systems* 17 (2004), pp. 1121–1128.

[111] Laura Walker Renninger, Preeti Verghese, and James Coughlan. "Where to look next? Eye movements reduce local uncertainty". en. In: *Journal of Vision* 7.3 (2007), p. 6. ISSN: 1534-7362. DOI: 10.1167/7.3.6.

[112] Jiri Najemnik and Wilson S Geisler. "Optimal eye movement strategies in visual search". en. In: *Nature* 139.6 (2005), pp. 1152–1153. DOI: 10.1038/nature03390.

[113] M. Wedel, R. Pieters, and R. van der Lans. "Modeling Eye Movements During Decision Making: A Review". In: *Psychometrika* 88.2 (2022), pp. 697–729. DOI: 10.1007/s11336-022-09876-4.

[114] A. Radulescu et al. "A Resource-Rational Account of Human Eye Movements During Immersive Visual Search". In: *Open Mind* 10 (Feb. 2026), pp. 91–117. ISSN: 2470-2986. DOI: 10.1162/OPMI.a.322.

[115] Vasily Tarasov. "General Fractional Dynamics". en. In: *Mathematics* 9.13 (2021), p. 1464. ISSN: 2227-7390. DOI: 10.3390/math9131464.

[116] Radford M. Neal. "Sampling from multimodal distributions using tempered transitions". en. In: *Statistics and Computing* 6.4 (1996), pp. 353–366. ISSN: 0960-3174, 1573-1375. DOI: 10.1007/BF00143556.

[117] K. Łatuszyński, Matthew Moores, and Timoth'ee Stumpf-F'etizon. *MCMC for multi-modal distributions*. en. 2026. DOI: 10.1201/9781003453420-14.

[118] Umut Simsekli. "Fractional Langevin Monte Carlo: Exploring Levy Driven Stochastic Differential Equations for Markov Chain Monte Carlo". In: *International Conference on Machine Learning* (2017).

[119] Ralf Metzler and Joseph Klafter. "The random walk's guide to anomalous diffusion: a fractional dynamics approach". en. In: *Physics Reports* 339.1 (2000), pp. 1–77. ISSN: 03701573. DOI: 10.1016/S0370-1573(00)00070-3.

[120] Julian Besag. *A Conceptual Introduction to Markov Chain Monte Carlo Methods*. en. 2004. DOI: 10.1007/978-1-4419-9017-4_11.

[121] Christophe Andrieu, Arnaud Doucet, and Roman Holenstein. "Particle Markov Chain Monte Carlo Methods". en. In: *Journal of the Royal Statistical Society Series B: Statistical Methodology* 72.3 (2010), pp. 269–342. ISSN: 1369-7412, 1467-9868. DOI: 10.1111/j.1467-9868.2009.00736.x.

[122] Michael Betancourt. *A Conceptual Introduction to Hamiltonian Monte Carlo*. en. 2017. DOI: 10.48550/arXiv.1701.02434.

[123] Nanyang Ye and Zhanxing Zhu. "Stochastic Fractional Hamiltonian Monte Carlo". In: *International Joint Conference on Artificial Intelligence* (July 2018), pp. 3019–3025. DOI: 10.24963/ijcai.2018/419.

[124] David Oechsler. "Lévy Langevin Monte Carlo". en. In: *Statistics and Computing* 34.1 (2024). ISSN: 0960-3174, 1573-1375. DOI: 10.1007/s11222-023-10345-w.

[125] Anita Behme and Claudius Lutke Schwienhorst. "Lévy Langevin Monte Carlo for sampling from heavy-tailed target distributions". en. In: *Extremes* (2025). ISSN: 1386-1999, 1572-915X. DOI: 10.1007/s10687-026-00544-9.

[126] Guozhang Chen and Pulin Gong. "A spatiotemporal mechanism of visual attention: Superdiffusive motion and theta oscillations of neural population activity patterns". en. In: *Science Advances* 8.16 (2022). ISSN: 2375-2548. DOI: 10.1126/sciadv.abl4995.

[127] Alessandro Barducci, Giovanni Bussi, and Michele Parrinello. "Well-Tempered Metadynamics: A Smoothly Converging and Tunable Free-Energy Method". en. In: *Physical Review Letters* 100.2 (2008). ISSN: 0031-9007, 1079-7114. DOI: 10.1103/PhysRevLett.100.020603.

[128] Giovanni Bussi and Alessandro Laio. "Using metadynamics to explore complex free-energy landscapes". en. In: *Nature Reviews Physics* 2.4 (2020), pp. 200–212. ISSN: 2522-5820. DOI: 10.1038/s42254-020-0153-0.

[129] Mark D. McDonnell, Shiro Ikeda, and Jonathan H. Manton. "An introductory review of information theory in the context of computational neuroscience". en. In: *Biological Cybernetics* 105.1 (2011), pp. 55–70. ISSN: 0340-1200, 1432-0770. DOI: 10.1007/s00422-011-0451-9.

[130] Yifei Liu and Rong Quan. "Effective Text-Directed Scanpath Prediction via Comprehensive Multi-modal Information Fusion". en. In: *Chinese Conference on Pattern Recognition and Computer Vision*. Ed. by Josef Kittler et al. Vol. 16280. Series Title: Lecture Notes in Computer Science. Singapore: Springer Nature Singapore, 2025, pp. 197–211. ISBN: 9789819557011 9789819557028. DOI: 10.1007/978-981-95-5702-8_14.

[131] E. Barkai. "Fractional Fokker-Planck equation, solution, and application". en. In: *Physical review. E, Statistical, nonlinear, and soft matter physics* 63.4 (2001). ISSN: 1063-651X, 1095-3787. DOI: 10.1103/PhysRevE.63.046118.

[132] Nicholas M Boffi and Eric Vanden-Eijnden. "Probability flow solution of the Fokker–Planck equation". en. In: *Machine Learning: Science and Technology* 4.3 (2022), p. 035012. ISSN: 2632-2153. DOI: 10.1088/2632-2153/ace2aa.

[133] Bhanuka Mahanama et al. "Eye Movement and Pupil Measures: A Review". en. In: *Frontiers of Computer Science* 3 (2022). ISSN: 2624-9898. DOI: 10.3389/fcomp.2021.733531.

[134] ERNST J. MEYER. "Rhythmicity of Saccadic Eye Movements". In: *A M A Archives of Ophthalmology* 76.4 (1966), pp. 554–556. ISSN: 0003-9950. DOI: 10.1001/archopht.1966.03850010556013.

[135] "Chapter 9 Saccadic Rhythm and Predictive Movement Sequences". In: *Advances in Psychology*. Ed. by Stephen Grossberg and Michael Kuperstein. Vol. 30. Advances in Psychology. Elsevier, 1986, pp. 203–226. DOI: 10.1016/s0166-4115(08)62516-9.

[136] Samson Chota et al. "Rhythmic fluctuations of saccadic reaction time arising from visual competition". In: *Scientific Reports* 8.1 (2018). ISSN: 2045-2322. DOI: 10.1038/s41598-018-34252-7.

[137] Yeganeh Shaverdi et al. "Orchestration of saccadic eye-movements by brain rhythms in macaque Frontal Eye Field". In: *Scientific Reports* 13.1 (2023). ISSN: 2045-2322. DOI: 10.1038/s41598-023-49346-0.

[138] Charles-Edouard Leroux et al. "The random walk of accommodation fluctuations". en. In: *Biomedical Optics Express* 12.11 (2021), p. 6897. ISSN: 2156-7085, 2156-7085. DOI: 10.1364/BOE.433926.

[139] James A. Roberts, Guy Wallis, and Michael Breakspear. "Fixational eye movements during viewing of dynamic natural scenes". en. In: *Frontiers in Psychology* 4 (2013). ISSN: 1664-1078. DOI: 10.3389/fpsyg.2013.00797.

[140] H. Collewijn and E. Kowler. "The significance of microsaccades for vision and oculomotor control". en. In: *Journal of Vision* 8.14 (2008), pp. 20–20. ISSN: 1534-7362. DOI: 10.1167/8.14.20.

[141] Jian-Qiao Zhu et al. "The autocorrelated Bayesian sampler: A rational process for probability judgments, estimates, confidence intervals, choices, confidence judgments, and response times." en. In: *Psychology Review* 131.2 (2023). ISSN: 1939-1471, 0033-295X. DOI: 10.1037/rev0000427.

[142] Nicholas J Butko and Javier R Movellan. "Infomax Control of Eye Movements". en. In: *IEEE Transactions on Autonomous Mental Development* 2.2 (2010), pp. 91–107. ISSN: 1943-0604, 1943-0612. DOI: 10.1109/TAMD.2010.2051029.

[143] Pratyush Tiwary and Michele Parrinello. "A Time-Independent Free Energy Estimator for Metadynamics". en. In: *Journal of Physical Chemistry B* 119.3 (2015), pp. 736–742. ISSN: 1520-6106, 1520-5207. DOI: 10.1021/jp504920s.

[144] Michela Ottobre et al. "A function space HMC algorithm with second order Langevin diffusion limit". In: *arXiv* 22.1 (2013), pp. 60–106. DOI: 10.3150/14-BEJ621.

[145] Ziming Liu, Andrew M. Stuart, and Yixuan Wang. *Second Order Ensemble Langevin Method for Sampling and Inverse Problems*. 2025. DOI: 10.48550/arXiv.2208.04506.

[146] Gonzalo Ruarte et al. "Integrating Ideal Bayesian Searcher and Neural Networks Models for Eye Movement Prediction in a Hybrid Search Task". In: *bioRxiv* (2024). DOI: 10.1101/2024.11.29.626088.

[147] Yunhui Zhou and Yuguo Yu. "Human visual search follows a suboptimal Bayesian strategy revealed by a spatiotemporal computational model and experiment". en. In: *Communications Biology* 4.1 (2021). ISSN: 2399-3642. DOI: 10.1038/s42003-020-01485-0.

[148] L. Itti, C. Koch, and E. Niebur. "A model of saliency-based visual attention for rapid scene analysis". en. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 20.11 (1998), pp. 1254–1259. ISSN: 01628828. DOI: 10.1109/34.730558.

[149] Surya Gayet and S"uheyla Aydemir. "Feature Integration Theory revisited: attention is not needed to bind stimulus features, but prevents them from falling apart". In: *Journal of Vision* 25.9 (2025), p. 1801. DOI: 10.1167/jov.25.9.1801.

[150] Jeremy M. Wolfe, Kyle R. Cave, and Susan L. Franzel. "Guided search: An alternative to the feature integration model for visual search." In: *Journal of Experimental Psychology: Human Perception and Performance* 15.3 (1989), pp. 419–433. ISSN: 1939-1277(Electronic),0096-1523(Print). DOI: 10.1037//0096-1523.15.3.419.

[151] Jeremy M. Wolfe. "Guided Search 6.0: An updated model of visual search". en. In: *Psychonomic Bulletin & Review* 28.4 (2021), pp. 1060–1092. ISSN: 1069-9384, 1531-5320. DOI: 10.3758/s13423-020-01859-9.